General Hospital Clinical Metadata and Workflow Implementation Manual

Edition 3.0 | 10 November 2026 | DHIS2 2.41, 2.42, 2.43 and 2.44

> **Deployment status:** Version-specific metadata payloads are structurally checked reference candidates; no successful live server import or clinical acceptance test has been performed. Retain a backup and run `importMode=VALIDATE` in staging before committing.

Clinical naming governance (edition 3.0)

All human-visible metadata names use concise clinical terminology without a product or implementation prefix. The names describe the clinical concept or business activity, while ownership and type are encoded in the stable `SSTTNNNN` metadata code and the registry. Examples: `Patient`, `Outpatient Consultation`, `Systolic Blood Pressure`, `Medication Administration`, `Laboratory Test Result`, and `Stock Receipt`.

Use recognised medical terminology and local clinical policy to review each concept before production. This catalogue has not been assigned verified SNOMED CT, LOINC, ICD-11, ATC, or UCUM codes automatically; external mappings must be clinically validated and licensed where applicable. Short names should remain readable in DHIS2 Capture. Avoid changing existing option value codes and preserve all 11-character metadata UIDs.

The accompanying `Metadata_Name_Changes.csv` records each exact UID, code, prior name and updated name. Codes and UIDs remain unchanged. Some similar clinical phrases may still refer to different measurement contexts; do not merge them without a semantic and dependency review.

1. Metadata coding specification

The approved format is `SSTTNNNN`, exactly eight uppercase alphanumeric characters. `SS` is the owner service or cross-cutting domain, `TT` is the two-letter metadata object type, and `NNNN` is a zero-padded four-digit sequence from 0001 through 9999. Codes never change when names change; sequences must not be recycled. An 11-character DHIS2 UID remains the authoritative reference.

Example code	Meaning
`OPPG0001`	Outpatient program
`IPPS0001`	Inpatient program stage
`VSDE0001`	Shared vital-sign data element
`PTTA0001`	Patient tracked entity attribute
`LBOS0001`	Laboratory option set
`OPPR0001`	Outpatient program rule

Example code	Meaning
OPPI0001	Outpatient program indicator

Metadata type dictionary: TE tracked entity type; TA tracked entity attribute; OS option set; OP option; DE data element; PG program; PS program stage; SC program stage section; PV program rule variable; PR program rule; PA program rule action; PI program indicator; RT relationship type.

Cross-cutting domain dictionary: GN general/shared, PT patient identity, VS vital signs, DX diagnosis, MD medicines, AL allergies, RF referral, WF workflow, BL billing, LB laboratory, AU audit, SC screening, HS clinical history. These are governed ownership prefixes rather than statements that a field is physically stored in every program.

Service ownership prefixes: OP Integrated Outpatient EMR, IP Integrated Inpatient EMR, IV Inventory and Supplies Tracker, ER Emergency and Resuscitation, AN Maternal Antenatal Care, MT Maternity Labour and Delivery, PN Postnatal and Newborn Follow-up, IM Child Health and Immunisation, HV HIV Care and ART, TB Tuberculosis and Leprosy Care, NC NCD Diabetes and Hypertension, ON Cancer and Oncology, MH Mental Health and Psychiatry, SU Surgical and Theatre Services, DN Dental and Oral Health, EY Eye and Vision Care, EN ENT Services, RH Rehabilitation and Physiotherapy, RN Renal Care and Dialysis, FP Family Planning and Reproductive Health, LB Laboratory Diagnostics, RD Imaging and Radiology, BB Blood Bank and Transfusion, NU Nutrition and Dietetics, EQ Facility Equipment and Maintenance, IC Hospital Infection Prevention and Control.

1.1 Cross-cutting metadata and reusable definitions

A data element or option set is classed as cross-cutting when referenced by multiple programs or owned by an explicitly shared clinical domain. Reuse is based on equivalent semantic definition, value type, unit, and option set, not on a similar label alone. See Metadata_Code_Registry.csv for each exact UID, code, classification and actual program use.

1.2 Option code preservation and patient data safety

Exception: DHIS2 option code values are clinical data values, so existing options retain their original DHIS2 codes. Each option also receives an eight-character **DER catalogue alias** (SSOPNNNN) stored in Option_Code_Alias_Map.json, but not substituted for the option code in import metadata. Otherwise historical data and rules could break. New option sets may adopt the DER code as the actual option code only before any values have been recorded and after integration review.

1.3 DHIS2 metadata references

All original UIDs are retained. Program, stage, variable, rule, indicator, data element, tracked entity type, tracked entity attribute, and option-set metadata receive an eight-character `code`. Foreign-key relationships remain UID-based. Option values are intentionally unchanged. Any external API or ETL using legacy metadata codes must be migrated deliberately, by UID lookup or the registry's previous-code mapping.

2. Package composition and counts

Metadata type	Count
trackedEntityTypes	3
trackedEntityAttributes	16
optionSets	61
options	316
dataElements	624
programs	26
programStages	163
programRuleVariables	8
programRules	8
programRuleActions	8
programIndicators	26

Total registry entries: **1259**. Cross-cutting registry entries: **172**. Preserved option data codes: **316**.

3. Program ownership and code index

Code	Program	Stage count
`OPPG0001`	Integrated Outpatient EMR	18
`IPPG0001`	Integrated Inpatient EMR	35
`IVPG0001`	Inventory and Supplies Tracker	13
`ERPG0001`	Emergency and Resuscitation	6
`ANPG0001`	Antenatal Care	5
`MTPG0001`	Labour and Delivery Care	6
`PNPG0001`	Postnatal and Newborn Follow-up	4
`IMPG0001`	Child Health and Immunisation	6
`HVPG0001`	HIV Care and Antiretroviral Therapy	5
`TBPG0001`	Tuberculosis and Leprosy Care	6
`NCPG0001`	Diabetes and Hypertension Care	5
`ONPG0001`	Cancer and Oncology	6
`MHPG0001`	Mental Health and Psychiatry	4

Code	Program	Stage count
SUPG0001	Surgical and Theatre Services	5
DNPG0001	Dental and Oral Health	3
EYPG0001	Eye and Vision Care	3
ENPG0001	Ear, Nose and Throat Services	3
RHPG0001	Rehabilitation and Physiotherapy	3
RNPG0001	Renal Care and Dialysis	3
FPPG0001	Family Planning and Reproductive Health	3
LBPG0001	Laboratory Diagnostics	5
RDPG0001	Imaging and Radiology	3
BBPG0001	Blood Bank and Transfusion	3
NUPG0001	Nutrition and Dietetics	3
EQPG0001	Facility Equipment and Maintenance	4
ICPG0001	Hospital Infection Prevention and Control	3

4. Service-point stage coding and workflow references

Each program stage is an event capture point, not a new tracked entity. The complete workflow specification follows this index. Linked medication orders, stock changes, payment transactions, and device handoffs remain transactionally controlled in their authoritative DER services.

Integrated Outpatient EMR (OPPG0001)

- OPPS0016 - **Registration Verification** (repeatable): Identity Verified, Registration Notes.
- OPPS0018 - **Visit Check-in** (repeatable): Visit Reference, Requested Service, Check-in Time, Queue Number, Visit Payment Category.
- OPPS0017 - **Triage and Vital Signs** (repeatable): Visit Reference, Triage Category, Temperature Celsius, Systolic Blood Pressure, Diastolic Blood Pressure, Heart Rate, Respiratory Rate, Oxygen Saturation Percent, Weight Kg, Height Cm.
- OPPS0004 - **Clinical Consultation** (repeatable): Visit Reference, Reason for Visit, History of Present Illness, Clinical Examination, Provisional Diagnosis, Final Diagnosis, Clinical Plan.
- OPPS0009 - **Laboratory Request** (repeatable): Visit Reference, Order Reference, Laboratory Test Requested, Order Priority, Specimen Type, Request Indication, Request Status.
- OPPS0011 - **Laboratory Specimen Collection** (repeatable): Visit Reference, Order Reference, Specimen Identifier, Collection Date Time, Specimen Received Date Time, Specimen Notes.
- OPPS0010 - **Laboratory Results** (repeatable): Visit Reference, Order Reference, Laboratory Test Name, Laboratory Result Value, Laboratory Result Interpretation, Laboratory Result Units, Laboratory Reference Range, Laboratory Report Date Time, Critical Result Escalated.

- `OPPS0008` - **Imaging Request** (repeatable): Visit Reference, Order Reference, Imaging Modality, Imaging Body Site, Imaging Clinical Indication, Order Priority.
- `OPPS0007` - **Imaging Report** (repeatable): Visit Reference, Order Reference, Imaging Findings, Imaging Impression, Imaging Performed Date Time.
- `OPPS0014` - **Procedure and Treatment** (repeatable): Visit Reference, Order Reference, Procedure Name, Procedure Date Time, Procedure Outcome, Adverse Event.
- `OPPS0013` - **Prescription** (repeatable): Visit Reference, Order Reference, Medicine Prescribed, Prescribed Dose, Dose Unit, Route of Administration, Frequency, Duration Days, Quantity Prescribed.
- `OPPS0012` - **Pharmacy Dispensing** (repeatable): Visit Reference, Order Reference, Medicine Dispensed, Quantity Dispensed, Dispensing Date Time, Dispensing Notes.
- `OPPS0001` - **Billing Charge** (repeatable): Visit Reference, Invoice Reference, Charge Line Reference, Charge Service Category, Charge Description, Charge Quantity, Charge Unit Price UGX, Charge Gross Amount UGX, Charge Discount Amount UGX, Charge Net Amount UGX.
- `OPPS0003` - **Billing Payment** (repeatable): Visit Reference, Invoice Reference, Payment Reference, Payment Method, Payment Amount UGX, Payment Date Time, Payment Status, External Transaction Reference, Payment Notes.
- `OPPS0002` - **Billing Clearance** (repeatable): Visit Reference, Invoice Reference, Billing Balance UGX, Billing Clearance Granted, Billing Clearance Reason, Billing Clearance Date Time.
- `OPPS0005` - **Clinical Review and Disposition** (repeatable): Visit Reference, Disposition, Clinical Outcome, Discharge Summary, Follow-up Date, Return Precautions.
- `OPPS0015` - **Referral and Transfer** (repeatable): Visit Reference, Referral Type, Referral Destination, Referral Reason, Referral Date, Referral Status.
- `OPPS0006` - **Follow-up Encounter** (repeatable): Visit Reference, Follow-up Findings, Clinical Outcome, Follow-up Date.

Integrated Inpatient EMR (`IPPG0001`)

- `IPPS0002` - **Admission Request and Bed Booking** (once per enrollment): Admission Request ID, Originating OPD Event ID, Proposed Admission Type, Requested Ward, Bed Request Time, Booking Outcome.
- `IPPS0003` - **Admission and Identity Verification** (once per enrollment): Admission ID, Admission Date and Time, Referred From, Admission Source, Reason for Admission, Admission Diagnosis, Admitting Clinician, Assigned Ward, Assigned Bed, Encounter ID.
- `IPPS0015` - **Initial Nursing Assessment** (repeatable): Encounter ID, Assessment Time, Presenting Concerns, Temperature Celsius, Pulse Per

Minute, Respiratory Rate Per Minute, Systolic BP, Diastolic BP, Oxygen Saturation Percent, Pain Score.

- `IPPS0001` - **Admission History and Physical** (repeatable): Encounter ID, History of Present Illness, Past Medical History, Surgical History, Allergy Description, Current Medicines, Physical Examination, Working Diagnosis, Care Plan, Clinical Priority.
- `IPPS0024` - **Patient Problems and Diagnoses** (repeatable): Encounter ID, Diagnosis Code, Diagnosis Description, Diagnosis Classification, Diagnosis Onset Date, Diagnosis Status, Responsible Clinician.
- `IPPS0035` - **Ward Round and Clinical Progress** (repeatable): Encounter ID, Round Time, Progress SOAP Subjective, Progress SOAP Objective, Progress SOAP Assessment, Progress SOAP Plan, Consultant ID, Workflow Status.
- `IPPS0033` - **Vital Signs and Early Warning** (repeatable): Encounter ID, Measurement Time, Temperature Celsius, Pulse Per Minute, Respiratory Rate Per Minute, Systolic BP, Diastolic BP, Oxygen Saturation Percent, Oxygen Delivery, Glasgow Coma Score.
- `IPPS0022` - **Medication Reconciliation** (repeatable): Encounter ID, Medication Name, Home Medication Dose, Home Medication Frequency, Medication Reconciliation Action, Reconciliation Reason, Reviewed By.
- `IPPS0021` - **Medication Prescription** (repeatable): Encounter ID, Prescription ID, Medication Order ID, Medicine Code, Medicine Name, Dose, Dose Unit, Route, Frequency, Start Date Time.
- `IPPS0026` - **Pharmacy Verification and Issue** (repeatable): Encounter ID, Prescription ID, Medication Order ID, Dispensing ID, Medicine Code, Quantity Issued, Batch Number, Expiry Date, Pharmacist ID, Workflow Status.
- `IPPS0020` - **Medication Administration Record** (repeatable): Encounter ID, Medication Order ID, Scheduled Dose Time, Actual Administration Time, Dose Administered, Dose Unit, Route, Administration Outcome, Omission Reason, Administered By.
- `IPPS0017` - **Intravenous Fluids and Infusions** (repeatable): Encounter ID, Infusion Order ID, Fluid Type, Total Volume ML, Infusion Rate ML Per Hour, Start Date Time, End Date Time, Administration Outcome, Recorded By.
- `IPPS0018` - **Laboratory Investigation Request** (repeatable): Encounter ID, Laboratory Request ID, Test Order ID, Test Code, Test Name, Sample Type, Clinical Indication, Clinical Priority, Request Time, Workflow Status.
- `IPPS0031` - **Specimen Collection and Handover** (repeatable): Encounter ID, Laboratory Request ID, Specimen ID, Collection Time, Sample Type, Collector ID, Specimen Condition, Handover Time, Workflow Status.
- `IPPS0019` - **Laboratory Results and Validation** (repeatable): Encounter ID, Laboratory Request ID, Test Order ID, Specimen ID, Test Code, Result Value, Result Unit, Reference Range, Result Interpretation,

Result Time.

- `IPPS0013` - **Imaging and Diagnostics Request** (repeatable): Encounter ID, Imaging Request ID, Imaging Modality, Body Region, Clinical Indication, Clinical Priority, Request Time, Workflow Status.
- `IPPS0012` - **Imaging Result and Report** (repeatable): Encounter ID, Imaging Request ID, Study Reference, Study Date Time, Report Findings, Report Impression, Reporting Clinician, Critical Result Alert, Workflow Status.
- `IPPS0029` - **Procedure and Intervention** (repeatable): Encounter ID, Procedure ID, Procedure Code, Procedure Name, Procedure Date Time, Consent Status, Procedure Notes, Procedure Outcome, Procedure Clinician, Complications.
- `IPPS0032` - **Surgery and Operating Theatre** (repeatable): Encounter ID, Surgery ID, Surgical Procedure, Preoperative Diagnosis, Surgery Start Time, Surgery End Time, Surgeon ID, Anesthesia Type, Safety Checklist Complete, Operation Notes.
- `IPPS0027` - **Post Anaesthesia and Recovery** (repeatable): Encounter ID, Surgery ID, Recovery Start Time, Recovery Score, Airway Status, Pain Score, Complications, Recovery Disposition, Receiving Staff.
- `IPPS0023` - **Nursing Care Plan and Interventions** (repeatable): Encounter ID, Nursing Problem, Nursing Intervention, Care Goal, Intervention Date Time, Response to Care, Responsible Nurse, Workflow Status.
- `IPPS0016` - **Intake Output and Fluid Balance** (repeatable): Encounter ID, Observation Time, Intake Type, Intake Volume ML, Output Type, Output Volume ML, Shift, Fluid Balance Note.
- `IPPS0006` - **Blood and Transfusion** (repeatable): Encounter ID, Transfusion Order ID, Blood Product Code, Blood Unit Identifier, Compatibility Result, Transfusion Start, Transfusion End, Transfusion Outcome, Adverse Reaction.
- `IPPS0007` - **Consultation and Specialty Review** (repeatable): Encounter ID, Consultation Request ID, Specialty, Reason for Consultation, Consultation Date Time, Clinical Opinion, Recommendations, Consultant ID, Workflow Status.
- `IPPS0034` - **Ward Bed Transfer and Handover** (repeatable): Encounter ID, Transfer ID, From Ward, From Bed, To Ward, To Bed, Transfer Reason, Transfer Time, Handover Summary, Receiving Staff.
- `IPPS0008` - **Critical Care and Escalation** (repeatable): Encounter ID, Escalation Time, Escalation Trigger, Escalation Level, Critical Care Plan, Responsible Clinician, Workflow Status.
- `IPPS0004` - **Allied Health and Rehabilitation** (repeatable): Encounter ID, Service Category, Assessment Summary, Intervention, Session Date Time, Progress Note, Follow Up Required.
- `IPPS0014` - **Infection Prevention and Isolation** (repeatable): Encounter ID, Infection Risk, Isolation Type, Isolation Start, Isolation End, Infection Control Action, Infection Control Review.
- `IPPS0005` - **Billing Charge and Claims** (repeatable): Encounter ID,

Charge ID, Charge Source Event ID, Charge Item Code, Charge Amount, Currency Code, Payer Type, Claim Reference, Payment Status.

- `IPPS0025` **- Payment and Financial Clearance** (repeatable): Encounter ID, Invoice ID, Payment Reference, Payment Amount, Currency Code, Payment Method, Payment Status, Waiver or Exemption, Financial Clearance Status.
- `IPPS0010` **- Discharge Planning and Education** (repeatable): Encounter ID, Estimated Discharge Date, Discharge Education, Home Medicines, Follow Up Appointment, Discharge Barriers, Caregiver Instructions, Workflow Status.
- `IPPS0011` **- Discharge Summary and Disposition** (once per enrollment): Encounter ID, Discharge Date Time, Final Diagnosis, Secondary Diagnoses, Treatment Summary, Procedures Summary, Discharge Medicines, Discharge Condition, Disposition, Follow Up Plan.
- `IPPS0030` **- Referral Transfer Out and Continuity** (repeatable): Encounter ID, Referral ID, Receiving Facility, Referral Reason, Referral Priority, Transfer Time, Receiving Contact, Handover Completed, Referral Status.
- `IPPS0009` **- Death and Mortuary Handover** (once per enrollment): Encounter ID, Date Time of Death, Death Certification Reference, Cause of Death Summary, Certifying Clinician, Mortuary Transfer Reference, Family Notified.
- `IPPS0028` **- Post Discharge Follow-up and Readmission** (repeatable): Encounter ID, Follow Up Date, Follow Up Outcome, Complication Identified, Readmission Reference, Follow Up Plan.

Inventory and Supplies Tracker (`IVPG0001`)

- `IVPS0006` **- INV - Opening Balance and Stocktake** (repeatable): INV - Opening Quantity, INV - Count Date, INV - Stock Location, INV - Batch or Lot Number, INV - Expiry Date, INV - Count Reference, INV - Approved By.
- `IVPS0004` **- INV - Goods Receipt** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0005` **- INV - Issue to Service Point** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0002` **- INV - Department Receipt** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.

- `IVPS0007` **- INV - Patient Dispensing** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0001` **- INV - Clinical Consumption** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0009` **- INV - Return and Quarantine** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0013` **- INV - Stock Transfer Dispatch** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0003` **- INV - Expiry Damage and Disposal** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0010` **- INV - Stock Adjustment** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0012` **- INV - Stock Reservation** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0011` **- INV - Stock Requisition** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.
- `IVPS0008` **- INV - Procurement and Replenishment** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID.

Emergency and Resuscitation (`ERPG0001`)

- `ERPS0001` **- Arrival and Emergency Registration** (repeatable): Arrival mechanism, Urgency, Presenting complaint, Encounter reference.

- `ERPS0005` - **Emergency Triage** (repeatable): Clinical acuity, Systolic BP (mmHg), Heart rate (per min), SpO2 (%), Red flag finding.
- `ERPS0006` - **Primary Survey and Resuscitation** (repeatable): Airway status, Breathing status, Circulation status, Resuscitation intervention, Time of intervention.
- `ERPS0002` - **Emergency Consultation and Orders** (repeatable): Clinical note, Diagnosis code, Order reference, Prescription reference.
- `ERPS0004` - **Emergency Procedures and Observation** (repeatable): Procedure code, Procedure status, Clinical result, Observation note.
- `ERPS0003` - **Emergency Disposition and Handover** (repeatable): Encounter disposition, Referral reference, Handover recipient, Handover time.

Antenatal Care (`ANPG0001`)

- `ANPS0003` - **ANC Registration and Pregnancy Dating** (once per enrollment): Last menstrual period, Estimated delivery date, Gravidity, Parity, Pregnancy risk history.
- `ANPS0005` - **ANC Visit and Risk Assessment** (repeatable): Gestational age weeks, Systolic BP (mmHg), Diastolic BP (mmHg), Weight (kg), Danger signs.
- `ANPS0004` - **ANC Screening and Diagnostics** (repeatable): HIV test result, Syphilis screening, Haemoglobin (g/dL), Ultrasound order reference, Clinical result.
- `ANPS0001` - **ANC Preventive Care and Counselling** (repeatable): Preventive intervention, Supplement issued, Counselling note, Follow-up appointment.
- `ANPS0002` - **ANC Referral and Outcome** (repeatable): Referral reference, Pregnancy outcome, Outcome date.

Labour and Delivery Care (`MTPG0001`)

- `MTPS0002` - **Labour Admission** (once per enrollment): Gestational age weeks, Labour onset time, Maternal risk note.
- `MTPS0003` - **Labour Monitoring** (repeatable): Cervical dilatation cm, Fetal heart rate, Contraction frequency, Systolic BP (mmHg).
- `MTPS0005` - **Obstetric Intervention** (repeatable): Procedure code, Intervention indication, Procedure status.
- `MTPS0001` - **Delivery and Birth Record** (repeatable): Birth mode, Birth date and time, Birth outcome, Birth weight (g), Newborn linked patient identifier.
- `MTPS0006` - **Postpartum Immediate Care** (repeatable): Estimated blood loss (mL), Postpartum complication, Maternal disposition.
- `MTPS0004` - **Maternity Discharge and Handover** (once per enrollment): Discharge date, Follow-up appointment, Referral reference.

Postnatal and Newborn Follow-up (PNPG0001)

- PNPS0003 - **Postnatal Enrollment** (once per enrollment): Delivery date, Delivery facility, Mother linked patient identifier.
- PNPS0004 - **Postnatal Maternal Assessment** (repeatable): Systolic BP (mmHg), Danger signs, Breastfeeding counselling.
- PNPS0001 - **Newborn Assessment** (repeatable): Birth weight (g), Weight (kg), Newborn danger signs, Jaundice assessment.
- PNPS0002 - **Postnatal Counselling and Referral** (repeatable): Counselling note, Referral reference, Follow-up status.

Child Health and Immunisation (IMPG0001)

- IMPS0003 - **Child Registration and Eligibility** (once per enrollment): Caregiver identifier, Birth date verified, Vaccination card reference.
- IMPS0004 - **Growth and Nutrition Assessment** (repeatable): Weight (kg), Height (cm), MUAC (mm), Nutrition classification.
- IMPS0005 - **Immunisation Due Assessment** (repeatable): Vaccine antigen code, Dose sequence, Vaccination status, Due date.
- IMPS0006 - **Vaccination Administration** (repeatable): Vaccine antigen code, Dose sequence, Vaccine lot number, Administration date, Vaccinator identifier.
- IMPS0001 - **Adverse Event Following Immunisation** (repeatable): AEFI onset, AEFI severity, AEFI clinical description, Referral reference.
- IMPS0002 - **Child Clinical Review and Referral** (repeatable): Clinical note, Diagnosis code, Referral reference.

HIV Care and Antiretroviral Therapy (HVPG0001)

- HVPS0004 - **HIV Enrollment and Baseline** (once per enrollment): Confirmed diagnosis date, ART start date, Baseline clinical stage, Treatment regimen code.
- HVPS0002 - **HIV Clinical Follow-up** (repeatable): Weight (kg), Clinical note, Treatment adherence, Next review date.
- HVPS0005 - **HIV Laboratory Monitoring** (repeatable): Viral load copies/mL, CD4 count, Specimen reference, Clinical result.
- HVPS0001 - **ART Prescription and Dispensing** (repeatable): Prescription reference, Treatment regimen code, Quantity, Stock transaction reference.
- HVPS0003 - **HIV Differentiated Service and Referral** (repeatable): Service delivery model, Referral reference, Follow-up status.

Tuberculosis and Leprosy Care (TBPG0001)

- TBPS0003 - **TB/Leprosy Screening and Classification** (once per enrollment): Disease classification, Symptoms duration days, Diagnosis code, Index case reference.

- `TBPS0002` - **Microbiology and Diagnostic Workup** (repeatable): Specimen reference, Molecular test result, Smear result.
- `TBPS0005` - **Treatment Initiation** (once per enrollment): Treatment start date, Treatment regimen code, Treatment phase.
- `TBPS0004` - **Treatment Adherence and Follow-up** (repeatable): Treatment adherence, Weight (kg), Treatment adverse effect.
- `TBPS0001` - **Contact Tracing and Referral** (repeatable): Contact patient identifier, Screening result, Referral reference.
- `TBPS0006` - **Treatment Outcome** (once per enrollment): TB outcome, Outcome date.

Diabetes and Hypertension Care (`NCPG0001`)

- `NCPS0003` - **NCD Enrollment and Risk Profile** (once per enrollment): Diabetes type, Hypertension diagnosis, Cardiovascular risk factors.
- `NCPS0004` - **NCD Review and Vitals** (repeatable): Systolic BP (mmHg), Diastolic BP (mmHg), Weight (kg), Clinical note.
- `NCPS0001` - **Diabetes Laboratory Monitoring** (repeatable): HbA1c (%), Fasting glucose (mmol/L), Urine albumin result.
- `NCPS0002` - **NCD Complications Screening** (repeatable): Foot exam result, Retinal screening result, Renal risk note.
- `NCPS0005` - **NCD Treatment and Review** (repeatable): Prescription reference, Treatment adherence, Follow-up appointment.

Cancer and Oncology (`ONPG0001`)

- `ONPS0002` - **Oncology Registration and Diagnosis** (once per enrollment): Primary cancer site code, Histology terminology code, Diagnosis date, Stage at diagnosis.
- `ONPS0001` - **Multidisciplinary Tumour Board** (repeatable): Decision date, Clinical note, Care plan status.
- `ONPS0006` - **Systemic Therapy Plan and Cycle** (repeatable): Protocol code, Cycle number, Treatment administration date, Treatment adverse effect.
- `ONPS0005` - **Surgery and Radiation Coordination** (repeatable): Procedure code, Radiotherapy course reference, Procedure status.
- `ONPS0003` - **Oncology Response and Survivorship** (repeatable): Treatment response code, Clinical note, Follow-up appointment.
- `ONPS0004` - **Palliative Oncology and Outcome** (repeatable): Pain score (0-10), Palliative intervention, Outcome date.

Mental Health and Psychiatry (`MHPG0001`)

- `MHPS0001` - **Mental Health Intake and Consent** (once per enrollment): Presenting concern, Consent status, Mental health risk.
- `MHPS0003` - **Mental Status and Risk Assessment** (repeatable): Mental state examination, Suicide risk note, Mental health risk.

- `MHPS0002` - **Mental Health Treatment Planning** (repeatable): Diagnosis code, Care plan status, Prescription reference.
- `MHPS0004` - **Psychotherapy and Follow-up** (repeatable): Therapy session summary, Follow-up status, Referral reference.

Surgical and Theatre Services (`SUPG0001`)

- `SUPS0004` - **Surgical Booking and Consent** (once per enrollment): Planned procedure code, Surgery date, Consent status.
- `SUPS0003` - **Preoperative Assessment** (repeatable): Anaesthesia risk classification, Preoperative findings, Procedure status.
- `SUPS0005` - **Theatre Safety Checklist** (repeatable): Sign-in completed, Time-out completed, Sign-out completed.
- `SUPS0001` - **Operative Record** (repeatable): Procedure code, Procedure start time, Procedure end time, Operative note, Wound classification.
- `SUPS0002` - **Postoperative and Surgical Review** (repeatable): Pain score (0-10), Postoperative complication, Follow-up appointment.

Dental and Oral Health (`DNPG0001`)

- `DNPS0001` - **Dental Assessment** (repeatable): Tooth notation system, Tooth identifier, Dental finding code, Pain score (0-10).
- `DNPS0003` - **Dental Procedure and Treatment** (repeatable): Tooth identifier, Procedure code, Procedure status, Prescription reference.
- `DNPS0002` - **Dental Follow-up** (repeatable): Clinical note, Follow-up appointment.

Eye and Vision Care (`EYPG0001`)

- `EYPS0003` - **Ophthalmology Screening** (repeatable): Visual acuity left, Visual acuity right, Presenting symptom.
- `EYPS0002` - **Ophthalmology Examination** (repeatable): Intraocular pressure left, Intraocular pressure right, Eye finding code.
- `EYPS0001` - **Eye Procedure and Referral** (repeatable): Procedure code, Referral reference, Follow-up appointment.

Ear, Nose and Throat Services (`ENPG0001`)

- `ENPS0001` - **ENT Assessment** (repeatable): ENT chief complaint, Otoscopy finding, Hearing screening result.
- `ENPS0002` - **ENT Investigation and Procedure** (repeatable): Order reference, Procedure code, Clinical result.
- `ENPS0003` - **ENT Review and Referral** (repeatable): Clinical note, Referral reference.

Rehabilitation and Physiotherapy (RHPG0001)

- RHPS0001 - **Rehabilitation Intake** (once per enrollment): Functional baseline scale, Therapy goal.
- RHPS0003 - **Rehabilitation Session** (repeatable): Therapy intervention code, Session duration minutes, Functional outcome score.
- RHPS0002 - **Rehabilitation Reassessment and Discharge** (repeatable): Functional outcome score, Discharge recommendation.

Renal Care and Dialysis (RNPG0001)

- RNPS0003 - **Renal Assessment and Access** (once per enrollment): Renal diagnosis code, Vascular access type, Baseline renal function note.
- RNPS0002 - **Dialysis Session** (repeatable): Dialysis session status, Pre dialysis weight kg, Post dialysis weight kg, Dialysis duration minutes.
- RNPS0001 - **Dialysis Complication and Review** (repeatable): Complication note, Referral reference, Follow-up appointment.

Family Planning and Reproductive Health (FPPG0001)

- FPPS0002 - **Family Planning Counselling** (repeatable): Method preference, Counselling note, Consent status.
- FPPS0001 - **Contraceptive Service** (repeatable): Contraceptive method code, Quantity, Procedure code, Service date.
- FPPS0003 - **Family Planning Follow-up** (repeatable): Adverse effect note, Follow-up status.

Laboratory Diagnostics (LBPG0001)

- LBPS0002 - **Laboratory Order Intake** (repeatable): Order reference, Encounter reference, Test catalogue code, Specimen reference, Order status.
- LBPS0005 - **Specimen Collection and Accession** (repeatable): Specimen reference, Specimen status, Collection date and time.
- LBPS0004 - **Laboratory Test Processing** (repeatable): Test catalogue code, Analyzer reference, Quality control result, Clinical result.
- LBPS0003 - **Laboratory Results and Validation** (repeatable): Order reference, Result value text, Result units, Reference interval, Result authorised by.
- LBPS0001 - **Laboratory Critical Result Communication** (repeatable): Critical result alert, Notified clinician ID, Notification time.

Imaging and Radiology (RDPG0001)

- RDPS0003 - **Imaging Request and Scheduling** (repeatable): Order reference, Modality code, Body site code, Appointment date.

- `RDPS0001` - **Imaging Acquisition** (repeatable): Imaging study identifier, Contrast administered, Procedure status.
- `RDPS0002` - **Imaging Reporting and Communication** (repeatable): Imaging study identifier, Radiology report, Clinical result, Reported date.

Blood Bank and Transfusion (`BBPG0001`)

- `BBPS0003` - **Transfusion Request and Eligibility** (repeatable): Order reference, Blood component code, Units requested, Clinical indication.
- `BBPS0001` - **Crossmatch and Component Issue** (repeatable): Blood unit identifier, Blood compatibility, Issue date and time.
- `BBPS0002` - **Transfusion Administration and Monitoring** (repeatable): Blood unit identifier, Start date and time, End date and time, Transfusion reaction.

Nutrition and Dietetics (`NUPG0001`)

- `NUPS0003` - **Nutrition Screening** (repeatable): Weight (kg), Height (cm), MUAC (mm), Nutrition classification.
- `NUPS0002` - **Nutrition Plan and Counselling** (repeatable): Diet plan reference, Counselling note, Care plan status.
- `NUPS0001` - **Nutrition Monitoring** (repeatable): Weight (kg), Nutrition response, Follow-up appointment.

Facility Equipment and Maintenance (`EQPG0001`)

- `EQPS0003` - **Equipment Registration** (once per enrollment): Asset identifier, Equipment category, Manufacturer, Serial number, Equipment location.
- `EQPS0004` - **Preventive Maintenance** (repeatable): Maintenance date, Maintenance outcome, Next maintenance date.
- `EQPS0001` - **Equipment Breakdown and Repair** (repeatable): Fault reported at, Repair action, Equipment return to service.
- `EQPS0002` - **Equipment Decommissioning** (once per enrollment): Decommission date, Decommission reason.

Hospital Infection Prevention and Control (`ICPG0001`)

- `ICPS0002` - **IPC Incident Notification** (repeatable): Incident reference, Incident date, IPC event classification, Service point.
- `ICPS0003` - **IPC Investigation and Measures** (repeatable): Investigation findings, Intervention action, Care plan status.
- `ICPS0001` - **IPC Closure and Review** (repeatable): Closure date, Lessons learned.

5. Metadata code and integration acceptance checklist

1. Export current DHIS2 metadata and patient data backup; verify none of the candidate UIDs collides with unrelated objects.
2. Review `Metadata_Code_Registry.csv` for external code dependencies, and compare all pre-existing codes with third-party mappings.
3. Apply shared foundation first, then programs/stages, then program rules/indicators; only proceed when each import report has no critical errors.
4. Validate program rules, rule variables, rule actions, option codes, and program indicators against the target DHIS2 release in a staging instance.
5. Validate clinical data values, units, permissible value ranges, terminology catalogue and the configured national health reporting package.
6. Confirm users, organisation units, sharing configuration and program access levels.
7. Run offline and WAN recovery scenarios, with event/relationship reconciliation and duplicate financial and stock transaction prevention.
8. Reconcile clinical indicators against authoritative hospital registers before production activation.

6. Complete hospital workflow documentation (preserved from edition 1)

DER General Hospital: Integrated DHIS2 Tracker and Operational Architecture

Reference implementation specification | Revision 1.0 | 10 November 2026

Scope: General hospital with OPD, emergency, IPD, maternity, specialised clinics, diagnostic services, pharmacy, inventory, billing, and governance. **Targets:** DHIS2 2.41, 2.42, 2.43 and 2.44. **Status:** locally generated reference metadata; no successful live metadata import or program-rule validation has been performed.

> **Clinical safety and deployment boundary:** This is an implementation baseline, not a certified EMR, a national clinical terminology package, a complete prescribing engine, or approval to run clinical care. Clinical protocols, drug catalogues, controlled vocabularies, national reporting requirements, patient consent and regulated workflows must be validated and approved locally before deployment.

1. Architecture, authority and data boundaries

A single **shared Patient Tracked Entity Type** and stable patient identifier are used across OPD, IPD, emergency and specialty programs. One patient can

have separate enrollments in different programs. OPD is longitudinal; IPD is episode-based (one enrollment per admission). Emergency and referral pathways may launch linked encounters. A new program enrollment is **not** a new patient.

Layer	DHIS2 representation	Authoritative service
Identity	Patient tracked entity + attributes	Master patient index / trusted identity governance
Clinical programs	Tracker programs, stages, events	Configured clinical capture and approved record source
Visit or admission	OPD encounter reference; IPD enrollment	DER encounter coordinator
Orders and fulfilment	Order/result event references and eligible native relationships	DER order engine or LIS/RIS/pharmacy system
Inventory	Commodity Tracker reporting plus movement references	DER inventory ledger or designated eLMIS
Billing	Clinical charge/payment summary events	Financial ledger, payment gateway or HIS
Reporting	Program indicators and aggregate reporting	DHIS2 analytics and nationally approved HMIS indicators
Files and images	Identifiers or metadata references	PACS, DMS or secure attachment storage

Identity and data-model constraints

- Do not create an unrelated Patient TE type for every clinic. The Inventory package retains a separate commodity TE; Equipment uses an equipment TE. These are not persons.
- Reuse existing patient identifiers and, after MPI review, link duplicate candidates through a controlled merge workflow; do not auto-merge by name.
- Separate **patient**, **program enrollment**, **visit/encounter**, **service event**, **order**, **result**, **prescription**, **administration**, **dispensing**, **stock movement** and **charge/payment** identities.
- Use reference identifiers inside event fields only for retrieval and interoperability. A text reference alone is **not** a DHIS2 native relationship. Configure eligible native relationshipTypes on the target instance and mirror operational references in DER.

- Multi-test orders and multi-item prescriptions are collections in a transactional DER/LIS/pharmacy model; Tracker repeatable events are derived interoperable representations where appropriate.

2. Hospital patient pathways

2.1 OPD pathway

Patient identity search -> visit check-in -> triage/vitals -> consultation -> parallel laboratory, imaging, procedures, pharmacy and billing -> clinician review -> disposition/referral/follow-up. Billing charges may arise at each service point; emergency care must not be blocked for payment.

2.2 Emergency pathway

Arrival -> acuity triage -> resuscitation if indicated -> clinical decision and orders -> observation or procedure -> home / urgent referral / hospital admission / transfer. Emergent life-saving care bypasses routine payment clearance.

2.3 IPD pathway

Decision to admit -> bed request and admission -> initial physician/nursing assessment -> daily rounds/medication administration/orders/procedures and multidisciplinary reviews -> transfers and intensive care where needed -> discharge medication/billing/summary/follow-up. A new IPD enrollment is created for a new admission, not for every ward round.

2.4 Specialty pathways

Each specialty has its own longitudinal enrollment when care requires specific longitudinal context (HIV, NCD, oncology, TB, ANC) or encounter-cycle enrollment when episodic care requires separate episodes (surgery, emergency, childbirth). Internal stages serve as service-point milestones, not independent copies of the patient.

3. Program catalogue and stage-by-stage workflows

The lists below specify purpose and data collection points for every generated specialty program. Existing OPD/IPD/inventory stage details remain in their accompanying specifications, preserved and referenced in the package. Actual clinical decisions must use approved clinical guidance.

Emergency and Resuscitation

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. Arrival and Emergency Registration Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Arrival mechanism, Urgency, Presenting complaint, Encounter reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Emergency Triage Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Clinical acuity, Systolic BP (mmHg), Heart rate (per min), SpO2 (%), Red flag finding. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Primary Survey and Resuscitation Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Airway status, Breathing status, Circulation status, Resuscitation intervention, Time of intervention. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Emergency Consultation and Orders Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Clinical note, Diagnosis code, Order reference, Prescription reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Emergency Procedures and Observation Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Procedure code, Procedure status, Clinical result, Observation note. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

6. Emergency Disposition and Handover Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Encounter disposition, Referral reference, Handover recipient, Handover time. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Maternal Antenatal Care

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. ANC Registration and Pregnancy Dating Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order

context; capture Last menstrual period, Estimated delivery date, Gravidity, Parity, Pregnancy risk history. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. ANC Visit and Risk Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Gestational age weeks, Systolic BP (mmHg), Diastolic BP (mmHg), Weight (kg), Danger signs. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. ANC Screening and Diagnostics Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture HIV test result, Syphilis screening, Haemoglobin (g/dL), Ultrasound order reference, Clinical result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. ANC Preventive Care and Counselling Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Preventive intervention, Supplement issued, Counselling note, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. ANC Referral and Outcome Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Referral reference, Pregnancy outcome, Outcome date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Maternity Labour and Delivery

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. Labour Admission Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Gestational age weeks, Labour onset time, Maternal risk note. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Labour Monitoring Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Cervical dilatation cm, Fetal heart rate, Contraction frequency, Systolic BP (mmHg). Record responsible

user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Obstetric Intervention Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Procedure code, Intervention indication, Procedure status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Delivery and Birth Record Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Birth mode, Birth date and time, Birth outcome, Birth weight (g), Newborn linked patient identifier. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Postpartum Immediate Care Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Estimated blood loss (mL), Postpartum complication, Maternal disposition. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

6. Maternity Discharge and Handover Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Discharge date, Follow-up appointment, Referral reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Postnatal and Newborn Follow-up

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. Postnatal Enrollment Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Delivery date, Delivery facility, Mother linked patient identifier. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Postnatal Maternal Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Systolic BP (mmHg), Danger signs, Breastfeeding counselling. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER

operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Newborn Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Birth weight (g), Weight (kg), Newborn danger signs, Jaundice assessment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Postnatal Counselling and Referral Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Counselling note, Referral reference, Follow-up status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Child Health and Immunisation

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Child Registration and Eligibility Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Caregiver identifier, Birth date verified, Vaccination card reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Growth and Nutrition Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Weight (kg), Height (cm), MUAC (mm), Nutrition classification. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Immunisation Due Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Vaccine antigen code, Dose sequence, Vaccination status, Due date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Vaccination Administration Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Vaccine antigen code, Dose sequence, Vaccine lot number, Administration date, Vaccinator identifier. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Adverse Event Following Immunisation Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture AEFI onset, AEFI severity, AEFI clinical description, Referral reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

6. Child Clinical Review and Referral Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Clinical note, Diagnosis code, Referral reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

HIV Care and ART

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. HIV Enrollment and Baseline Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Confirmed diagnosis date, ART start date, Baseline clinical stage, Treatment regimen code. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. HIV Clinical Follow-up Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Weight (kg), Clinical note, Treatment adherence, Next review date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. HIV Laboratory Monitoring Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Viral load copies/mL, CD4 count, Specimen reference, Clinical result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. ART Prescription and Dispensing Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Prescription reference, Treatment regimen code, Quantity, Stock transaction reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. HIV Differentiated Service and Referral Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture

Service delivery model, Referral reference, Follow-up status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Tuberculosis and Leprosy Care

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. TB/Leprosy Screening and Classification Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Disease classification, Symptoms duration days, Diagnosis code, Index case reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Microbiology and Diagnostic Workup Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Specimen reference, Molecular test result, Smear result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Treatment Initiation Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Treatment start date, Treatment regimen code, Treatment phase. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Treatment Adherence and Follow-up Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Treatment adherence, Weight (kg), Treatment adverse effect. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Contact Tracing and Referral Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Contact patient identifier, Screening result, Referral reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

6. Treatment Outcome Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture TB outcome, Outcome date. Record responsible user, timestamp, source service

point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

NCD Diabetes and Hypertension

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. NCD Enrollment and Risk Profile Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Diabetes type, Hypertension diagnosis, Cardiovascular risk factors. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. NCD Review and Vitals Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Systolic BP (mmHg), Diastolic BP (mmHg), Weight (kg), Clinical note. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Diabetes Laboratory Monitoring Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture HbA1c (%), Fasting glucose (mmol/L), Urine albumin result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. NCD Complications Screening Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Foot exam result, Retinal screening result, Renal risk note. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. NCD Treatment and Review Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Prescription reference, Treatment adherence, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Cancer and Oncology

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. Oncology Registration and Diagnosis Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Primary cancer site code, Histology terminology code, Diagnosis date, Stage at diagnosis. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Multidisciplinary Tumour Board Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Decision date, Clinical note, Care plan status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Systemic Therapy Plan and Cycle Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Protocol code, Cycle number, Treatment administration date, Treatment adverse effect. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Surgery and Radiation Coordination Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Procedure code, Radiotherapy course reference, Procedure status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Oncology Response and Survivorship Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Treatment response code, Clinical note, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

6. Palliative Oncology and Outcome Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Pain score (0-10), Palliative intervention, Outcome date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Mental Health and Psychiatry

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Mental Health Intake and Consent Typically once per enrollment. Ser-

vice point responsibilities: identify and verify the encounter or order context; capture Presenting concern, Consent status, Mental health risk. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Mental Status and Risk Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Mental state examination, Suicide risk note, Mental health risk. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Mental Health Treatment Planning Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Diagnosis code, Care plan status, Prescription reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Psychotherapy and Follow-up Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Therapy session summary, Follow-up status, Referral reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Surgical and Theatre Services

Enrollment policy: repeatable episode-based enrollment. **Tracked entity:** shared patient.

1. Surgical Booking and Consent Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Planned procedure code, Surgery date, Consent status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Preoperative Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Anaesthesia risk classification, Preoperative findings, Procedure status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Theatre Safety Checklist Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Sign-in completed, Time-out completed, Sign-out completed. Record responsible user, timestamp, source

service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Operative Record Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Procedure code, Procedure start time, Procedure end time, Operative note, Wound classification. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Postoperative and Surgical Review Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Pain score (0-10), Postoperative complication, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Dental and Oral Health

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Dental Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Tooth notation system, Tooth identifier, Dental finding code, Pain score (0-10). Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Dental Procedure and Treatment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Tooth identifier, Procedure code, Procedure status, Prescription reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Dental Follow-up Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Clinical note, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Eye and Vision Care

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Ophthalmology Screening Repeatable. Service point responsibilities:

identify and verify the encounter or order context; capture Visual acuity left, Visual acuity right, Presenting symptom. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Ophthalmology Examination Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Intraocular pressure left, Intraocular pressure right, Eye finding code. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Eye Procedure and Referral Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Procedure code, Referral reference, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

ENT Services

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. ENT Assessment Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture ENT chief complaint, Otoscopy finding, Hearing screening result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. ENT Investigation and Procedure Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Order reference, Procedure code, Clinical result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. ENT Review and Referral Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Clinical note, Referral reference. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Rehabilitation and Physiotherapy

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Rehabilitation Intake Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Functional baseline scale, Therapy goal. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Rehabilitation Session Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Therapy intervention code, Session duration minutes, Functional outcome score. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Rehabilitation Reassessment and Discharge Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Functional outcome score, Discharge recommendation. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Renal Care and Dialysis

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Renal Assessment and Access Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Renal diagnosis code, Vascular access type, Baseline renal function note. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Dialysis Session Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Dialysis session status, Pre dialysis weight kg, Post dialysis weight kg, Dialysis duration minutes. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Dialysis Complication and Review Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Complication note, Referral reference, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Family Planning and Reproductive Health

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Family Planning Counselling Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Method preference, Counselling note, Consent status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Contraceptive Service Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Contraceptive method code, Quantity, Procedure code, Service date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Family Planning Follow-up Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Adverse effect note, Follow-up status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Laboratory Diagnostics

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Laboratory Order Intake Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Order reference, Encounter reference, Test catalogue code, Specimen reference, Order status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Specimen Collection and Accession Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Specimen reference, Specimen status, Collection date and time. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Laboratory Test Processing Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Test catalogue code, Analyzer reference, Quality control result, Clinical result. Record responsible user, timestamp, source service point, relevant consent and onward handoff as

DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Laboratory Results and Validation Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Order reference, Result value text, Result units, Reference interval, Result authorised by. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

5. Laboratory Critical Result Communication Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Critical result alert, Notified clinician ID, Notification time. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Imaging and Radiology

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Imaging Request and Scheduling Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Order reference, Modality code, Body site code, Appointment date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Imaging Acquisition Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Imaging study identifier, Contrast administered, Procedure status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Imaging Reporting and Communication Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Imaging study identifier, Radiology report, Clinical result, Reported date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Blood Bank and Transfusion

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Transfusion Request and Eligibility Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Order reference, Blood component code, Units requested, Clinical indication. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Crossmatch and Component Issue Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Blood unit identifier, Blood compatibility, Issue date and time. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Transfusion Administration and Monitoring Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Blood unit identifier, Start date and time, End date and time, Transfusion reaction. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Nutrition and Dietetics

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. Nutrition Screening Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Weight (kg), Height (cm), MUAC (mm), Nutrition classification. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Nutrition Plan and Counselling Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Diet plan reference, Counselling note, Care plan status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Nutrition Monitoring Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Weight (kg), Nutrition response, Follow-up appointment. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Facility Equipment and Maintenance

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** equipment.

1. Equipment Registration Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Asset identifier, Equipment category, Manufacturer, Serial number, Equipment location. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. Preventive Maintenance Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Maintenance date, Maintenance outcome, Next maintenance date. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

3. Equipment Breakdown and Repair Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Fault reported at, Repair action, Equipment return to service. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Equipment Decommissioning Typically once per enrollment. Service point responsibilities: identify and verify the encounter or order context; capture Decommission date, Decommission reason. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

Hospital Infection Prevention and Control

Enrollment policy: longitudinal / single enrollment (baseline; subject to clinical review). **Tracked entity:** shared patient.

1. IPC Incident Notification Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Incident reference, Incident date, IPC event classification, Service point. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

2. IPC Investigation and Measures Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Investigation findings, Intervention action, Care plan status. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER

operational context. Check identity, required values, configured access rights and local approvals before completion.

3. IPC Closure and Review Repeatable. Service point responsibilities: identify and verify the encounter or order context; capture Closure date, Lessons learned. Record responsible user, timestamp, source service point, relevant consent and onward handoff as DER operational context. Check identity, required values, configured access rights and local approvals before completion.

4. Cross-cutting order, result, medication and stock workflows

Lab: consultation/ward request -> one order with one or more tests -> specimen accession -> individual observations/results -> verification -> critical-value notification -> clinician acknowledgment. Preserve specimen and order identifiers; do not place multiple analytes in a single unstructured result field.

Imaging: request -> scheduling -> study acquisition/PACS storage -> report -> clinical review. DICOM data remains in PACS; only identifiers and selected report content belong in Tracker.

Medication: prescription item(s) -> pharmacist clinical verification -> reservation -> dispensing (lot and quantity) -> medication administration record when IPD -> stock deduction exactly once -> patient record and billing update. Prescribing, dispensing and administration are distinct events. Maintain allergy checks, interactions and dosing alerts in a governed medication engine; generic sample metadata is not sufficient for medication safety.

Blood products: request -> consent/eligibility -> component reservation and crossmatch -> issue -> two-person bedside verification -> administration/monitoring -> traceability and haemovigilance. Include national rules for blood donation and transfusion identifiers.

Stock: purchases/receipts -> put-away -> transfer -> available-to-promise -> issue/consumption/dispensing -> reconciliation, returns, quarantine, expiry and recall. Ledger posting is transactional and idempotent. Clients with no connection to a common DER server must not independently decrement shared stock without controlled allocations.

Billing: capture itemised charge-source references, exemptions, payers, refunds and reconciliation within a dedicated financial service. DHIS2 summary events are informational; do not treat them as an authoritative double-entry financial ledger.

5. DHIS2 program relationships and DER associations

Relationship	Scope	Integration strategy
Mother–newborn	Patient to/from patient	Native TE–TE when configured; consent and child-protection restrictions
OPD consultation–IPD admission	Event to/from enrollment	Configured native relation where supported, plus encounter transition mapping
Request–result	Event to/from event	Native only where constraints allow; operational order IDs otherwise
Prescription–dispense–administration	Event to/from event	Validate eligible stage constraints; DER lifecycle owns each item
Patient–contact	Patient to/from patient	Controlled contact tracing and consent/authorisation
Service-point handoff	DER task to/from event	DER-only workflow link, never falsely exported as native
Commodity movement–dispensing	Inventory ledger to/from clinical event	Reference and idempotency mapping, not a patient-to-stock relationship

6. Metadata components and counts

Metadata type	Count
trackedEntityTypes	3
trackedEntityAttributes	16
optionSets	61
options	316
dataElements	624
programs	26
programStages	163
programRuleVariables	8
programRules	8
programRuleActions	8
programIndicators	26

6.1 Option sets

Each of the following locally authored option sets contains named values and stable option codes. The sets are NOT authoritative ICD, LOINC, SNOMED, ATC or WHO package imports. - **Visit Status:** Scheduled, Arrived, In progress, Completed, Cancelled, Did not attend. - **Referral Priority:** Routine, Urgent, Emergency. - **Encounter Disposition:** Home, Admitted, Transferred, Referred, Left against advice, Deceased. - **Clinical Acuity:** Immediate, Very urgent, Urgent, Standard, Non urgent. - **Order Status:** Requested, Accepted, In progress, Completed, Cancelled, Rejected. - **Clinical Result:** Normal, Abnormal, Critical, Indeterminate, Not performed. - **Care Plan Status:** Planned, Active, On hold, Completed, Cancelled. - **Treatment Adherence:** Good, Partial, Poor, Unknown. - **TB Treatment Outcome:** Cured, Treatment completed, Treatment failed, Died, Lost to follow up, Not evaluated. - **Pregnancy Outcome:** Live birth, Stillbirth, Miscarriage, Termination, Ectopic, Unknown. - **Birth Mode:** Spontaneous vaginal, Assisted vaginal, Caesarean. - **Vaccination Status:** Due, Administered, Deferred, Contraindicated, Refused. - **Wound Classification:** Clean, Clean-contaminated, Contaminated, Dirty. - **Mental Health Risk:** Low, Moderate, High, Imminent. - **Screening Outcome:** Negative, Positive, Indeterminate, Not done. - **Nutrition Classification:** Normal, Moderate acute malnutrition, Severe acute malnutrition, Overweight, Obesity. - **Procedure Status:** Planned, Started, Completed, Abandoned, Cancelled. - **Renal Session Status:** Planned, Active, Completed, Interrupted, Cancelled. - **Blood Compatibility:** Compatible, Incompatible, Not determined. - **Follow-up Status:** Due, Completed, Missed, Rescheduled. - **Payment Status:** Unbilled, Pending, Partially paid, Paid, Exempt, Deferred. - **Consent Status:** Given, Declined, Withdrawn, Not applicable. - **Yes No Unknown:** Yes, No, Unknown. - **Specimen Status:** Requested, Collected, Received, Rejected, Processed, Reported.

6.2 Program rules and indicators

The package contains 8 deliberately limited, event-local warning rules, linked program rule variables and actions. Their expressions use current-event numeric fields and **must** be validated through the target DHIS2 API because rule evaluation and available action semantics vary across versions and applications.

There is one event-count indicator per included program (26 total). These are operational baselines, not WHO-standard coverage or clinical quality indicators. Numerator/denominator indicators (e.g. ANC retention, viral suppression, TB outcomes, maternal mortality, bed occupancy, stock-out rate) require approved case definitions, denominator populations, code lists and analytics validation before release.

6.3 Examples of locally approved indicator definitions

Indicator	Numerator	Denominator	Safeguards
Inpatient mortality proportion	IPD discharge episodes with death disposition	All completed IPD discharge episodes	Case-mix and referral interpretation required
ANC first-contact coverage	First ANC contacts among resident pregnancies	Estimated eligible pregnancies in catchment	National ANC definition and population source
HIV viral suppression	Patients with latest eligible viral load below approved threshold	Eligible ART clients with reported viral load	Use national guidelines; missing tests handled explicitly
TB treatment success	Cured + treatment completed cohort	Defined treatment-start cohort	Cohort closure and transfer-out rules
Stock-out days	Days with zero available usable quantity	Days product expected in stock	Stock location, reporting completeness
Emergency median door-to-triage	Arrival-to-triage minutes per eligible visit	Eligible emergency arrivals	Compute in DER analytics; not generic event count
Bed occupancy	Occupied bed-days	Available staffed bed-days	Requires operational bed census, not only Tracker events

7. International standards and terminology governance

- **WHO ICD-11**: classification for diagnoses where adopted by national policy. Use an authoritative versioned release and validated code service; do not manufacture clinical diagnosis option lists.
- **SNOMED CT**: detailed clinical concepts where licensing/usage arrangements permit. Map coded diagnoses, findings, procedures and problems through a terminology service.
- **LOINC**: laboratory test and observation identifiers; use controlled units and reference ranges appropriate to the laboratory and test method.
- **UCUM**: standard units of measure and explicit conversions; prevent dose/unit confusion.
- **WHO ATC/DDD or national essential medicines catalogue**: medication classification/catalogue mapping; use appropriate medicine product identifiers and national formulary as separate governed data.

- **HL7 FHIR R4/R5 profiles as nationally adopted**: Patient, Encounter, Condition, Observation, ServiceRequest, Specimen, DiagnosticReport, MedicationRequest, MedicationDispense, MedicationAdministration, Procedure, Immunization, Appointment, Claim/Invoice (as profiled). Map semantics; DHIS2 does not automatically implement all these resources.
- **DICOM**: imaging assets remain in imaging infrastructure, referenced by accession/study identifiers.
- **WHO SMART Guidelines / DHIS2 health data toolkit**: reconcile programme indicators with national health information standards, not placeholder definitions.
- **Data protections**: patient consent, RBAC/ABAC, purpose limitation, sensitive clinic segmentation, auditing, retention policies, encryption, backup and restoration and breach response must be configured for Ugandan law and hospital governance.

8. Version profiles: DHIS2 2.41, 2.42, 2.43, 2.44

The four JSON files intentionally use **one conservative common metadata schema and stable identifiers**. They are separate labelled import candidates for each version, not four independently tested exports. Newer features should be introduced only after target-server metadata inspection, expression validation and dry-run import. Do not mix alternate-version bundles on the same server.

Target	Package	Required gate
2.41	`DER_General_Hospital_DHIS2_2_41_Metadata.json`	Dry-run import, program-rule description validation, Tracker capture tests, permissions, analytics
2.42	`DER_General_Hospital_DHIS2_2_42_Metadata.json`	Dry-run import, program-rule description validation, Tracker capture tests, permissions, analytics
2.43	`DER_General_Hospital_DHIS2_2_43_Metadata.json`	Dry-run import, program-rule description validation, Tracker capture tests, permissions, analytics

Target	Package	Required gate
2.44	`DER_General_Hospital_DHIS2_2_44_Metadata.json`	DHIS2 import, program-rule description validation, Tracker capture tests, permissions, analytics

Recommended import order for split layers: shared catalogue -> programs and program stages -> rule variables/rules/actions and indicators. Treat partial imports cautiously; referential integrity and permissions are server-owned. The complete version JSON is the reference bundle.

Compatibility test protocol

1. Back up the DHIS2 database; use a staging clone. Confirm exact DHIS2 patch version and installed applications.
2. GET existing `/api/metadata` (or targeted `/api/programs`, `/api/trackedEntityTypes`, `/api/dataElements`, `/api/optionSets`) and compare UIDs, codes, names, conflicts, categories and ownership.
3. First use `POST /api/metadata?importMode=VALIDATE&importStrategy=CREATE` (or version-supported dry-run endpoint/parameters); capture complete import report. Exact parameters must be checked against server version.
4. Resolve duplicate objects, invalid fields, option-set constraints and program stage associations; do not modify existing national metadata blindly.
5. Validate every program-rule condition and action expression through the instance's expression-description endpoints, and check evaluated rules in Capture and Tracker importer.
6. Verify clinical data fixtures against `/api/tracker` with `importMode=VALIDATE`; test TE, enrollment, event and eligible relationship references, including offline replay.
7. Rebuild analytics and validate program-indicator expressions and reported data against independent hand-calculated fixtures.
8. Configure organisation unit assignments, program access levels, program stage sharing, data protection, patient search permissions, user groups and service-point roles.
9. Test clinical scenarios and safety conditions with locally authorised clinicians and administrators before any production deployment.

9. DER Server and Client requirements

Client: App launcher (Capture / Patient Search / Emergency / OPD / IPD / Specialist Clinics / Laboratory / Pharmacy / Inventory / Billing / Administration), contextual work queues, dedicated list and detail pages, metadata-driven forms with clear sections, patient timeline, clinical tasks, related-record drawer,

accessible tablets/desktop, structured search, autosave drafts, and offline status. Do not place a full list and a full data-entry form on the same page.

Server: federated patient identity mapping, site-level shared records, authorization and consent checks, order lifecycle, durable outbox/inbox journal, idempotency keys, optimistic concurrency, conflict classification, central DHIS2 interoperability, local-to-central UID mapping, terminology service, attachment references, central sync observability, transactional inventory and financial adapters.

Cross-service workflow: clinician signs a lab order -> lab queue shows the order -> collector records specimen accession -> analyser/result reviewer finalises test -> clinician sees and acknowledges result. Other services run in parallel; the client shows clear responsibility and next step without duplicating identifiers.

Safety: emergency care never blocked by billing; critical values notified and acknowledged; unsynchronised edits clearly visible; clinical conflicts require role-authorised resolution; offline medicine and stock changes are serialised by a site ledger or allocated quotas; LWW is prohibited for clinical or stock conflicts.

10. Roles, minimum permissions and audit

Role	Default scope	Sensitive controls
Registration	Identity lookup/verification and visit initiation	No unrestricted clinical access
Nurse/midwife	Triage, nursing, maternity forms	Stage- and facility-scoped
Clinician	Assessment, diagnosis, orders and discharge	Signed records; reasoned amendments
Lab technologist	Orders, accession and validation	No unrelated patient chart browsing
Radiologist	Study/report access	PACS and restricted reports
Pharmacist	Verification, dispense and stock	Separation from prescribing where needed
Stock officer	Receipts, transfers, counts	No clinical diagnosis disclosure
Billing officer	Charges, payment reconciliation	No unrelated confidential clinical content
Administrator	Metadata/users/operations	No automatic clinical content access
Auditor	Authorised read-only trails	Immutable logs and export approvals

11. Testing, known limitations and acceptance criteria

- **Static validation:** UID syntax, uniqueness, internal references, option-set codes, stage-field references, variable-field references, reproducibility and version manifest consistency.
- **Integration validation:** all four target DHIS2 versions require their own actual import tests; no such tests are represented as complete here.
- **Clinical fixtures:** patient repeat visits; multi-encounter IPD; ANC-to-delivery; maternal/newborn links; child vaccine doses; TB regimen outcomes; HIV viral load; oncology cycles; urgent lab results; pharmacy partial fills; stock adjustments; emergency admission; financial reconciliation.
- **Offline:** WAN outage and restoration, client restart, duplicate event replay, concurrent changes across devices, metadata drift, rejected central edits, patient merge and recovery.
- **Reporting:** indicators validated with reference calculations and data access controls; no derived indicator should be treated as nationally approved by default.
- **Explicit limitations:** this metadata does not bundle exhaustive coded clinical vocabularies, therapeutic dose rules, DICOM binary assets, actual national reporting datasets, bed capacity ledger, or financial accounting. These require separately governed integration modules.

12. Source references and implementation further reading

- DHIS2 metadata API, 2.41: https://docs.dhis2.org/en/develop/using-the-api/dhis-core-version-241/metadata.html
- DHIS2 metadata API, 2.42: https://docs.dhis2.org/en/develop/using-the-api/dhis-core-version-242/metadata.html
- DHIS2 Tracker API: https://docs.dhis2.org/en/develop/using-the-api/dhis-core-version-master/tracker.html
- DHIS2 health data toolkit: https://dhis2.org/health-data-toolkit/
- DHIS2 and FHIR interoperability: https://dhis2.org/integration/fhir/
- WHO ICD-11: https://icd.who.int/
- LOINC: https://loinc.org/
- SNOMED International: https://www.snomed.org/
- HL7 FHIR: https://hl7.org/fhir/
- UCUM: https://ucum.org/

Appendix A. Existing source programmes

Integrated Outpatient EMR

- **Registration Verification** (repeatable): Identity Verified, Registration Notes.
- **Visit Check-in** (repeatable): Visit Reference, Requested Service, Check-in Time, Queue Number, Visit Payment Category.

- **Triage and Vital Signs** (repeatable): Visit Reference, Triage Category, Temperature Celsius, Systolic Blood Pressure, Diastolic Blood Pressure, Heart Rate, Respiratory Rate, Oxygen Saturation Percent, Weight Kg, Height Cm, Triage Complaint.
- **Clinical Consultation** (repeatable): Visit Reference, Reason for Visit, History of Present Illness, Clinical Examination, Provisional Diagnosis, Final Diagnosis, Clinical Plan.
- **Laboratory Request** (repeatable): Visit Reference, Order Reference, Laboratory Test Requested, Order Priority, Specimen Type, Request Indication, Request Status.
- **Laboratory Specimen Collection** (repeatable): Visit Reference, Order Reference, Specimen Identifier, Collection Date Time, Specimen Received Date Time, Specimen Notes.
- **Laboratory Results** (repeatable): Visit Reference, Order Reference, Laboratory Test Name, Laboratory Result Value, Laboratory Result Interpretation, Laboratory Result Units, Laboratory Reference Range, Laboratory Report Date Time, Critical Result Escalated.
- **Imaging Request** (repeatable): Visit Reference, Order Reference, Imaging Modality, Imaging Body Site, Imaging Clinical Indication, Order Priority.
- **Imaging Report** (repeatable): Visit Reference, Order Reference, Imaging Findings, Imaging Impression, Imaging Performed Date Time.
- **Procedure and Treatment** (repeatable): Visit Reference, Order Reference, Procedure Name, Procedure Date Time, Procedure Outcome, Adverse Event.
- **Prescription** (repeatable): Visit Reference, Order Reference, Medicine Prescribed, Prescribed Dose, Dose Unit, Route of Administration, Frequency, Duration Days, Quantity Prescribed.
- **Pharmacy Dispensing** (repeatable): Visit Reference, Order Reference, Medicine Dispensed, Quantity Dispensed, Dispensing Date Time, Dispensing Notes.
- **Billing Charge** (repeatable): Visit Reference, Invoice Reference, Charge Line Reference, Charge Service Category, Charge Description, Charge Quantity, Charge Unit Price UGX, Charge Gross Amount UGX, Charge Discount Amount UGX, Charge Net Amount UGX, Charge Source Reference, Payer Category, Charge Status.
- **Billing Payment** (repeatable): Visit Reference, Invoice Reference, Payment Reference, Payment Method, Payment Amount UGX, Payment Date Time, Payment Status, External Transaction Reference, Payment Notes.
- **Billing Clearance** (repeatable): Visit Reference, Invoice Reference, Billing Balance UGX, Billing Clearance Granted, Billing Clearance Reason, Billing Clearance Date Time.
- **Clinical Review and Disposition** (repeatable): Visit Reference, Disposition, Clinical Outcome, Discharge Summary, Follow-up Date, Return Precautions.
- **Referral and Transfer** (repeatable): Visit Reference, Referral Type,

Referral Destination, Referral Reason, Referral Date, Referral Status.

- **Follow-up Encounter** (repeatable): Visit Reference, Follow-up Findings, Clinical Outcome, Follow-up Date. ### Integrated Inpatient EMR
- **Admission Request and Bed Booking** (single): Admission Request ID, Originating OPD Event ID, Proposed Admission Type, Requested Ward, Bed Request Time, Booking Outcome.
- **Admission and Identity Verification** (single): Admission ID, Admission Date and Time, Referred From, Admission Source, Reason for Admission, Admission Diagnosis, Admitting Clinician, Assigned Ward, Assigned Bed, Encounter ID, Identity Verified, Consent Status, Next of Kin Contact.
- **Initial Nursing Assessment** (repeatable): Encounter ID, Assessment Time, Presenting Concerns, Temperature Celsius, Pulse Per Minute, Respiratory Rate Per Minute, Systolic BP, Diastolic BP, Oxygen Saturation Percent, Pain Score, Fall Risk, Pressure Injury Risk, Nursing Care Needs.
- **Admission History and Physical** (repeatable): Encounter ID, History of Present Illness, Past Medical History, Surgical History, Allergy Description, Current Medicines, Physical Examination, Working Diagnosis, Care Plan, Clinical Priority.
- **Patient Problems and Diagnoses** (repeatable): Encounter ID, Diagnosis Code, Diagnosis Description, Diagnosis Classification, Diagnosis Onset Date, Diagnosis Status, Responsible Clinician.
- **Ward Round and Clinical Progress** (repeatable): Encounter ID, Round Time, Progress SOAP Subjective, Progress SOAP Objective, Progress SOAP Assessment, Progress SOAP Plan, Consultant ID, Workflow Status.
- **Vital Signs and Early Warning** (repeatable): Encounter ID, Measurement Time, Temperature Celsius, Pulse Per Minute, Respiratory Rate Per Minute, Systolic BP, Diastolic BP, Oxygen Saturation Percent, Oxygen Delivery, Glasgow Coma Score, Early Warning Score, Escalation Action.
- **Medication Reconciliation** (repeatable): Encounter ID, Medication Name, Home Medication Dose, Home Medication Frequency, Medication Reconciliation Action, Reconciliation Reason, Reviewed By.
- **Medication Prescription** (repeatable): Encounter ID, Prescription ID, Medication Order ID, Medicine Code, Medicine Name, Dose, Dose Unit, Route, Frequency, Start Date Time, End Date Time, Quantity Ordered, Prescriber, Workflow Status.
- **Pharmacy Verification and Issue** (repeatable): Encounter ID, Prescription ID, Medication Order ID, Dispensing ID, Medicine Code, Quantity Issued, Batch Number, Expiry Date, Pharmacist ID, Workflow Status.
- **Medication Administration Record** (repeatable): Encounter ID, Medication Order ID, Scheduled Dose Time, Actual Administration Time, Dose Administered, Dose Unit, Route, Administration Outcome, Omission Reason, Administered By, Witnessed By.
- **Intravenous Fluids and Infusions** (repeatable): Encounter ID, Infusion Order ID, Fluid Type, Total Volume ML, Infusion Rate ML Per Hour,

Start Date Time, End Date Time, Administration Outcome, Recorded By.

- **Laboratory Investigation Request** (repeatable): Encounter ID, Laboratory Request ID, Test Order ID, Test Code, Test Name, Sample Type, Clinical Indication, Clinical Priority, Request Time, Workflow Status.
- **Specimen Collection and Handover** (repeatable): Encounter ID, Laboratory Request ID, Specimen ID, Collection Time, Sample Type, Collector ID, Specimen Condition, Handover Time, Workflow Status.
- **Laboratory Results and Validation** (repeatable): Encounter ID, Laboratory Request ID, Test Order ID, Specimen ID, Test Code, Result Value, Result Unit, Reference Range, Result Interpretation, Result Time, Verified By, Critical Result Alert.
- **Imaging and Diagnostics Request** (repeatable): Encounter ID, Imaging Request ID, Imaging Modality, Body Region, Clinical Indication, Clinical Priority, Request Time, Workflow Status.
- **Imaging Result and Report** (repeatable): Encounter ID, Imaging Request ID, Study Reference, Study Date Time, Report Findings, Report Impression, Reporting Clinician, Critical Result Alert, Workflow Status.
- **Procedure and Intervention** (repeatable): Encounter ID, Procedure ID, Procedure Code, Procedure Name, Procedure Date Time, Consent Status, Procedure Notes, Procedure Outcome, Procedure Clinician, Complications.
- **Surgery and Operating Theatre** (repeatable): Encounter ID, Surgery ID, Surgical Procedure, Preoperative Diagnosis, Surgery Start Time, Surgery End Time, Surgeon ID, Anesthesia Type, Safety Checklist Complete, Operation Notes, Postoperative Diagnosis.
- **Post Anaesthesia and Recovery** (repeatable): Encounter ID, Surgery ID, Recovery Start Time, Recovery Score, Airway Status, Pain Score, Complications, Recovery Disposition, Receiving Staff.
- **Nursing Care Plan and Interventions** (repeatable): Encounter ID, Nursing Problem, Nursing Intervention, Care Goal, Intervention Date Time, Response to Care, Responsible Nurse, Workflow Status.
- **Intake Output and Fluid Balance** (repeatable): Encounter ID, Observation Time, Intake Type, Intake Volume ML, Output Type, Output Volume ML, Shift, Fluid Balance Note.
- **Blood and Transfusion** (repeatable): Encounter ID, Transfusion Order ID, Blood Product Code, Blood Unit Identifier, Compatibility Result, Transfusion Start, Transfusion End, Transfusion Outcome, Adverse Reaction.
- **Consultation and Specialty Review** (repeatable): Encounter ID, Consultation Request ID, Specialty, Reason for Consultation, Consultation Date Time, Clinical Opinion, Recommendations, Consultant ID, Workflow Status.
- **Ward Bed Transfer and Handover** (repeatable): Encounter ID, Transfer ID, From Ward, From Bed, To Ward, To Bed, Transfer Reason, Transfer Time, Handover Summary, Receiving Staff.
- **Critical Care and Escalation** (repeatable): Encounter ID, Escalation

Time, Escalation Trigger, Escalation Level, Critical Care Plan, Responsible Clinician, Workflow Status.

- **Allied Health and Rehabilitation** (repeatable): Encounter ID, Service Category, Assessment Summary, Intervention, Session Date Time, Progress Note, Follow Up Required.
- **Infection Prevention and Isolation** (repeatable): Encounter ID, Infection Risk, Isolation Type, Isolation Start, Isolation End, Infection Control Action, Infection Control Review.
- **Billing Charge and Claims** (repeatable): Encounter ID, Charge ID, Charge Source Event ID, Charge Item Code, Charge Amount, Currency Code, Payer Type, Claim Reference, Payment Status.
- **Payment and Financial Clearance** (repeatable): Encounter ID, Invoice ID, Payment Reference, Payment Amount, Currency Code, Payment Method, Payment Status, Waiver or Exemption, Financial Clearance Status.
- **Discharge Planning and Education** (repeatable): Encounter ID, Estimated Discharge Date, Discharge Education, Home Medicines, Follow Up Appointment, Discharge Barriers, Caregiver Instructions, Workflow Status.
- **Discharge Summary and Disposition** (single): Encounter ID, Discharge Date Time, Final Diagnosis, Secondary Diagnoses, Treatment Summary, Procedures Summary, Discharge Medicines, Discharge Condition, Disposition, Follow Up Plan, Authorising Clinician.
- **Referral Transfer Out and Continuity** (repeatable): Encounter ID, Referral ID, Receiving Facility, Referral Reason, Referral Priority, Transfer Time, Receiving Contact, Handover Completed, Referral Status.
- **Death and Mortuary Handover** (single): Encounter ID, Date Time of Death, Death Certification Reference, Cause of Death Summary, Certifying Clinician, Mortuary Transfer Reference, Family Notified.
- **Post Discharge Follow-up and Readmission** (repeatable): Encounter ID, Follow Up Date, Follow Up Outcome, Complication Identified, Readmission Reference, Follow Up Plan. ### Inventory and Supplies Tracker
- **INV - Opening Balance and Stocktake** (repeatable): INV - Opening Quantity, INV - Count Date, INV - Stock Location, INV - Batch or Lot Number, INV - Expiry Date, INV - Count Reference, INV - Approved By.
- **INV - Goods Receipt** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Supplier Name, INV - Goods Received Note, INV - Purchase Order Reference, INV - Unit Cost.
- **INV - Issue to Service Point** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked

Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Destination Stock Location, INV - Requisition ID.

- **INV - Department Receipt** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Dispatch Reference, INV - Source Stock Location.
- **INV - Patient Dispensing** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Prescription ID, INV - Patient Reference Token.
- **INV - Clinical Consumption** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Service Event ID, INV - Consumption Context.
- **INV - Return and Quarantine** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Return Source, INV - Quality Assessment.
- **INV - Stock Transfer Dispatch** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Destination Stock Location, INV - Transfer Reference.
- **INV - Expiry Damage and Disposal** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Disposal Method, INV - Authorisation Reference.
- **INV - Stock Adjustment** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Counted Quantity, INV - System Quantity, INV - Adjust-

ment Approval.

- **INV - Stock Reservation** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Reservation ID, INV - Reserved Quantity, INV - Reservation Expiry.
- **INV - Stock Requisition** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Requisition ID, INV - Requested Quantity, INV - Requesting Service Point, INV - Required By.
- **INV - Procurement and Replenishment** (repeatable): INV - Transaction ID, INV - Source Operation ID, INV - Transaction Type, INV - Transaction Time, INV - Stock Location, INV - Quantity, INV - Base Unit, INV - Batch or Lot Number, INV - Expiry Date, INV - Linked Visit ID, INV - Linked Order ID, INV - Reason, INV - Performed By User ID, INV - Transaction Status, INV - Purchase Order Reference, INV - Supplier Name, INV - Ordered Quantity, INV - Expected Delivery.

Appendix B. Relationship, program rule and indicator setup checklist

1. Agree event to enrollment and event to event association constraints and direction with DHIS2 administrators before adding native relationshipTypes.
2. Confirm data elements, option set codes, choice labels, data value types, clinical thresholds and visibility before enabling rules.
3. Confirm calculations using program indicator validation API and sample patient cohorts; review cross-program indicators separately in analytics/warehouse.
4. Create distinct configuration profiles by facility and national jurisdiction; never overwrite authoritative national IDs or code systems.

7. Cross-cutting and service-specific code catalogue

The CSV/JSON registry is authoritative. The following index provides named fields and codes for rapid review.

trackedEntityTypes

DER Code	Metadata name	Scope
EQTE0001	Facility Equipment	service-specific
IVTE0001	Inventory Commodity	service-specific
PTTE0001	Patient	cross-cutting

trackedEntityAttributes

DER Code	Metadata name	Scope
PTTA0001	Date of Birth	cross-cutting
PTTA0002	Family Name	cross-cutting
PTTA0003	Given Name	cross-cutting
PTTA0004	Home Address	cross-cutting
PTTA0005	National Identifier	cross-cutting
PTTA0006	Patient Identifier	cross-cutting
PTTA0007	Phone Number	cross-cutting
PTTA0008	Sex	cross-cutting
PTTA0009	Inventory - Active Status	cross-cutting
PTTA0010	Inventory - Base Unit of Measure	cross-cutting
PTTA0011	Inventory - Commodity Code	cross-cutting
PTTA0012	Inventory - Commodity Description	cross-cutting
PTTA0013	Inventory - GTIN or Barcode	cross-cutting
PTTA0014	Inventory - Generic Name	cross-cutting
PTTA0015	Inventory - Product Category	cross-cutting
PTTA0016	Inventory - Strength or Specification	cross-cutting

optionSets

DER Code	Metadata name	Scope
BBOS0001	Blood Compatibility	service-specific
EROS0001	Clinical Acuity	service-specific
EROS0002	Referral Priority	service-specific
GNOS0001	Care Plan Status	cross-cutting
GNOS0002	Clinical Result	cross-cutting
GNOS0003	Consent Status	cross-cutting
GNOS0004	Encounter Disposition	cross-cutting
GNOS0005	Follow-up Status	cross-cutting
GNOS0006	Nutrition Classification	cross-cutting
GNOS0007	Payment Status	cross-cutting
GNOS0008	Pregnancy Outcome	cross-cutting
GNOS0009	Procedure Status	cross-cutting
GNOS0010	Treatment Adherence	cross-cutting
GNOS0011	Visit Status	cross-cutting

DER Code	Metadata name	Scope
GNOS0012	Yes No Unknown	cross-cutting
IMOS0001	Vaccination Status	service-specific
IPOS0001	IPD Administration Outcome	service-specific
IPOS0002	IPD Admission Source	service-specific
IPOS0003	IPD Consent Status	service-specific
IPOS0004	IPD Diagnosis Classification	service-specific
IPOS0005	IPD Diagnosis Status	service-specific
IPOS0006	IPD Disposition	service-specific
IPOS0007	IPD Escalation Level	service-specific
IPOS0008	IPD Financial Clearance Status	service-specific
IPOS0009	IPD Medication Reconciliation Action	service-specific
IPOS0010	IPD Payer Type	service-specific
IPOS0011	IPD Payment Method	service-specific
IPOS0012	IPD Payment Status	service-specific
IPOS0013	IPD Referral Status	service-specific
IPOS0014	IPD Route	service-specific
IPOS0015	IPD Shift	service-specific
IPOS0016	IPD Transfusion Outcome	service-specific
IPOS0017	IPD Workflow Status	service-specific
IVOS0001	INV Status	service-specific
IVOS0002	INV Stock Location	service-specific
IVOS0003	INV Stock Reason	service-specific
IVOS0004	INV Transaction Type	service-specific
LBOS0001	Order Status	cross-cutting
LBOS0002	Specimen Status	cross-cutting
MHOS0001	Mental Health Risk	service-specific
MTOS0001	Birth Mode	service-specific
OPOS0001	Clinical Outcome	service-specific
OPOS0002	Disposition	service-specific
OPOS0003	Invoice	service-specific
OPOS0004	Payer	service-specific
OPOS0005	Payment	service-specific
OPOS0006	Payment Status	service-specific
OPOS0007	Priority	service-specific
OPOS0008	Referral	service-specific
OPOS0009	Result	service-specific
OPOS0010	Service	service-specific
OPOS0011	Service Charge	service-specific
OPOS0012	Status	service-specific
OPOS0013	Triage	service-specific
OPOS0014	Unit	service-specific
OPOS0015	Yes/No	service-specific
PTOS0001	Sex	cross-cutting
RNOS0001	Renal Session Status	service-specific

DER Code	Metadata name	Scope
SCOS0001	Screening Outcome	cross-cutting
SUOS0001	Wound Classification	service-specific
TBOS0001	TB Treatment Outcome	service-specific

dataElements

DER Code	Metadata name	Scope
ANDE0001	Estimated delivery date	service-specific
ANDE0002	Gravidity	service-specific
ANDE0003	HIV test result	service-specific
ANDE0004	Haemoglobin (g/dL)	service-specific
ANDE0005	Last menstrual period	service-specific
ANDE0006	Parity	service-specific
ANDE0007	Pregnancy outcome	service-specific
ANDE0008	Pregnancy risk history	service-specific
ANDE0009	Preventive intervention	service-specific
ANDE0010	Supplement issued	service-specific
ANDE0011	Syphilis screening	service-specific
ANDE0012	Ultrasound order reference	service-specific
BBDE0001	Blood compatibility	service-specific
BBDE0002	Blood component code	service-specific
BBDE0003	Blood unit identifier	service-specific
BBDE0004	Clinical indication	service-specific
BBDE0005	End date and time	service-specific
BBDE0006	Issue date and time	service-specific
BBDE0007	Start date and time	service-specific
BBDE0008	Transfusion reaction	service-specific
BBDE0009	Units requested	service-specific
DNDE0001	Dental finding code	service-specific
DNDE0002	Tooth identifier	service-specific
DNDE0003	Tooth notation system	service-specific
DXDE0001	Diagnosis code	cross-cutting
ENDE0001	ENT chief complaint	service-specific
ENDE0002	Hearing screening result	service-specific
ENDE0003	Otoscopy finding	service-specific
EQDE0001	Asset identifier	service-specific
EQDE0002	Decommission date	service-specific
EQDE0003	Decommission reason	service-specific
EQDE0004	Equipment category	service-specific
EQDE0005	Equipment location	service-specific
EQDE0006	Equipment return to service	service-specific
EQDE0007	Fault reported at	service-specific

DER Code	Metadata name	Scope
EQDE0008	Maintenance date	service-specific
EQDE0009	Maintenance outcome	service-specific
EQDE0010	Manufacturer	service-specific
EQDE0011	Next maintenance date	service-specific
EQDE0012	Repair action	service-specific
EQDE0013	Serial number	service-specific
ERDE0001	Airway status	service-specific
ERDE0002	Arrival mechanism	service-specific
ERDE0003	Breathing status	service-specific
ERDE0004	Circulation status	service-specific
ERDE0005	Clinical acuity	service-specific
ERDE0006	Encounter disposition	service-specific
ERDE0007	Handover recipient	service-specific
ERDE0008	Handover time	service-specific
ERDE0009	Heart rate (per min)	service-specific
ERDE0010	Observation note	service-specific
ERDE0011	Presenting complaint	service-specific
ERDE0012	Red flag finding	service-specific
ERDE0013	Resuscitation intervention	service-specific
ERDE0014	SpO2 (%)	service-specific
ERDE0015	Time of intervention	service-specific
ERDE0016	Urgency	service-specific
EYDE0001	Eye finding code	service-specific
EYDE0002	Intraocular pressure left	service-specific
EYDE0003	Intraocular pressure right	service-specific
EYDE0004	Presenting symptom	service-specific
EYDE0005	Visual acuity left	service-specific
EYDE0006	Visual acuity right	service-specific
FPDE0001	Adverse effect note	service-specific
FPDE0002	Contraceptive method code	service-specific
FPDE0003	Method preference	service-specific
FPDE0004	Service date	service-specific
GNDE0001	Clinical Notes	cross-cutting
GNDE0002	Performed By	cross-cutting
GNDE0003	Service Point	cross-cutting
GNDE0004	Care plan status	cross-cutting
GNDE0005	Clinical note	cross-cutting
GNDE0006	Clinical result	cross-cutting
GNDE0007	Consent status	cross-cutting
GNDE0008	Cost (local currency)	cross-cutting
GNDE0009	Counselling note	cross-cutting
GNDE0010	Danger signs	cross-cutting
GNDE0011	Date and time	cross-cutting
GNDE0012	Diagnosis coding system	cross-cutting

DER Code	Metadata name	Scope
GNDE0013	Encounter date	cross-cutting
GNDE0014	Follow-up appointment	cross-cutting
GNDE0015	Follow-up status	cross-cutting
GNDE0016	Gestational age weeks	cross-cutting
GNDE0017	Invoice reference	cross-cutting
GNDE0018	Nutrition classification	cross-cutting
GNDE0019	Outcome date	cross-cutting
GNDE0020	Pain score (0-10)	cross-cutting
GNDE0021	Payment status	cross-cutting
GNDE0022	Procedure code	cross-cutting
GNDE0023	Procedure coding system	cross-cutting
GNDE0024	Procedure status	cross-cutting
GNDE0025	Quantity	cross-cutting
GNDE0026	Respiratory rate (per min)	cross-cutting
GNDE0027	Responsible clinician identifier	cross-cutting
GNDE0028	Temperature (C)	cross-cutting
GNDE0029	Treatment adherence	cross-cutting
GNDE0030	Treatment adverse effect	cross-cutting
GNDE0031	Workflow status	cross-cutting
HVDE0001	ART start date	service-specific
HVDE0002	Baseline clinical stage	service-specific
HVDE0003	CD4 count	service-specific
HVDE0004	Confirmed diagnosis date	service-specific
HVDE0005	Next review date	service-specific
HVDE0006	Service delivery model	service-specific
HVDE0007	Stock transaction reference	service-specific
HVDE0008	Viral load copies/mL	service-specific
ICDE0001	Closure date	service-specific
ICDE0002	IPC event classification	service-specific
ICDE0003	Incident date	service-specific
ICDE0004	Incident reference	service-specific
ICDE0005	Intervention action	service-specific
ICDE0006	Investigation findings	service-specific
ICDE0007	Lessons learned	service-specific
ICDE0008	Service point	service-specific
IMDE0001	AEFI clinical description	service-specific
IMDE0002	AEFI onset	service-specific
IMDE0003	AEFI severity	service-specific
IMDE0004	Administration date	service-specific
IMDE0005	Birth date verified	service-specific
IMDE0006	Caregiver identifier	service-specific
IMDE0007	Dose sequence	service-specific
IMDE0008	Due date	service-specific
IMDE0009	Vaccination card reference	service-specific

DER Code	Metadata name	Scope
IMDE0010	Vaccination status	service-specific
IMDE0011	Vaccinator identifier	service-specific
IMDE0012	Vaccine antigen code	service-specific
IMDE0013	Vaccine lot number	service-specific
IPDE0001	Actual Administration Time	service-specific
IPDE0002	Administered By	service-specific
IPDE0003	Administration Outcome	service-specific
IPDE0004	Admission Date and Time	service-specific
IPDE0005	Admission Diagnosis	service-specific
IPDE0006	Admission ID	service-specific
IPDE0007	Admission Request ID	service-specific
IPDE0008	Admission Source	service-specific
IPDE0009	Admitting Clinician	service-specific
IPDE0010	Adverse Reaction	service-specific
IPDE0011	Airway Status	service-specific
IPDE0012	Allergy Description	service-specific
IPDE0013	Anesthesia Type	service-specific
IPDE0014	Assessment Summary	service-specific
IPDE0015	Assessment Time	service-specific
IPDE0016	Assigned Bed	service-specific
IPDE0017	Assigned Ward	service-specific
IPDE0018	Authorising Clinician	service-specific
IPDE0019	Batch Number	service-specific
IPDE0020	Bed Request Time	service-specific
IPDE0021	Blood Product Code	service-specific
IPDE0022	Blood Unit Identifier	service-specific
IPDE0023	Body Region	service-specific
IPDE0024	Booking Outcome	service-specific
IPDE0025	Care Goal	service-specific
IPDE0026	Care Plan	service-specific
IPDE0027	Caregiver Instructions	service-specific
IPDE0028	Cause of Death Summary	service-specific
IPDE0029	Certifying Clinician	service-specific
IPDE0030	Charge Amount	service-specific
IPDE0031	Charge ID	service-specific
IPDE0032	Charge Item Code	service-specific
IPDE0033	Charge Source Event ID	service-specific
IPDE0034	Claim Reference	service-specific
IPDE0035	Clinical Indication	service-specific
IPDE0036	Clinical Opinion	service-specific
IPDE0037	Clinical Priority	service-specific
IPDE0038	Collection Time	service-specific
IPDE0039	Collector ID	service-specific
IPDE0040	Compatibility Result	service-specific

DER Code	Metadata name	Scope
IPDE0041	Complication Identified	service-specific
IPDE0042	Complications	service-specific
IPDE0043	Consent Status	service-specific
IPDE0044	Consultant ID	service-specific
IPDE0045	Consultation Date Time	service-specific
IPDE0046	Consultation Request ID	service-specific
IPDE0047	Critical Care Plan	service-specific
IPDE0048	Critical Result Alert	service-specific
IPDE0049	Currency Code	service-specific
IPDE0050	Current Medicines	service-specific
IPDE0051	Date Time of Death	service-specific
IPDE0052	Death Certification Reference	service-specific
IPDE0053	Diagnosis Classification	service-specific
IPDE0054	Diagnosis Code	service-specific
IPDE0055	Diagnosis Description	service-specific
IPDE0056	Diagnosis Onset Date	service-specific
IPDE0057	Diagnosis Status	service-specific
IPDE0058	Diastolic BP	service-specific
IPDE0059	Discharge Barriers	service-specific
IPDE0060	Discharge Condition	service-specific
IPDE0061	Discharge Date Time	service-specific
IPDE0062	Discharge Education	service-specific
IPDE0063	Discharge Medicines	service-specific
IPDE0064	Dispensing ID	service-specific
IPDE0065	Disposition	service-specific
IPDE0066	Dose	service-specific
IPDE0067	Dose Administered	service-specific
IPDE0068	Dose Unit	service-specific
IPDE0069	Early Warning Score	service-specific
IPDE0070	Encounter ID	service-specific
IPDE0071	End Date Time	service-specific
IPDE0072	Escalation Action	service-specific
IPDE0073	Escalation Level	service-specific
IPDE0074	Escalation Time	service-specific
IPDE0075	Escalation Trigger	service-specific
IPDE0076	Estimated Discharge Date	service-specific
IPDE0077	Expiry Date	service-specific
IPDE0078	Fall Risk	service-specific
IPDE0079	Family Notified	service-specific
IPDE0080	Final Diagnosis	service-specific
IPDE0081	Financial Clearance Status	service-specific
IPDE0082	Fluid Balance Note	service-specific
IPDE0083	Fluid Type	service-specific
IPDE0084	Follow Up Appointment	service-specific

DER Code	Metadata name	Scope
IPDE0085	Follow Up Date	service-specific
IPDE0086	Follow Up Outcome	service-specific
IPDE0087	Follow Up Plan	service-specific
IPDE0088	Follow Up Required	service-specific
IPDE0089	Frequency	service-specific
IPDE0090	From Bed	service-specific
IPDE0091	From Ward	service-specific
IPDE0092	Glasgow Coma Score	service-specific
IPDE0093	Handover Completed	service-specific
IPDE0094	Handover Summary	service-specific
IPDE0095	Handover Time	service-specific
IPDE0096	History of Present Illness	service-specific
IPDE0097	Home Medication Dose	service-specific
IPDE0098	Home Medication Frequency	service-specific
IPDE0099	Home Medicines	service-specific
IPDE0100	Identity Verified	service-specific
IPDE0101	Imaging Modality	service-specific
IPDE0102	Imaging Request ID	service-specific
IPDE0103	Infection Control Action	service-specific
IPDE0104	Infection Control Review	service-specific
IPDE0105	Infection Risk	service-specific
IPDE0106	Infusion Order ID	service-specific
IPDE0107	Infusion Rate ML Per Hour	service-specific
IPDE0108	Intake Type	service-specific
IPDE0109	Intake Volume ML	service-specific
IPDE0110	Intervention	service-specific
IPDE0111	Intervention Date Time	service-specific
IPDE0112	Invoice ID	service-specific
IPDE0113	Isolation End	service-specific
IPDE0114	Isolation Start	service-specific
IPDE0115	Isolation Type	service-specific
IPDE0116	Laboratory Request ID	service-specific
IPDE0117	Measurement Time	service-specific
IPDE0118	Medication Name	service-specific
IPDE0119	Medication Order ID	service-specific
IPDE0120	Medication Reconciliation Action	service-specific
IPDE0121	Medicine Code	service-specific
IPDE0122	Medicine Name	service-specific
IPDE0123	Mortuary Transfer Reference	service-specific
IPDE0124	Next of Kin Contact	service-specific
IPDE0125	Nursing Care Needs	service-specific
IPDE0126	Nursing Intervention	service-specific
IPDE0127	Nursing Problem	service-specific
IPDE0128	Observation Time	service-specific

DER Code	Metadata name	Scope
IPDE0129	Omission Reason	service-specific
IPDE0130	Operation Notes	service-specific
IPDE0131	Originating OPD Event ID	service-specific
IPDE0132	Output Type	service-specific
IPDE0133	Output Volume ML	service-specific
IPDE0134	Oxygen Delivery	service-specific
IPDE0135	Oxygen Saturation Percent	service-specific
IPDE0136	Pain Score	service-specific
IPDE0137	Past Medical History	service-specific
IPDE0138	Payer Type	service-specific
IPDE0139	Payment Amount	service-specific
IPDE0140	Payment Method	service-specific
IPDE0141	Payment Reference	service-specific
IPDE0142	Payment Status	service-specific
IPDE0143	Pharmacist ID	service-specific
IPDE0144	Physical Examination	service-specific
IPDE0145	Postoperative Diagnosis	service-specific
IPDE0146	Preoperative Diagnosis	service-specific
IPDE0147	Prescriber	service-specific
IPDE0148	Prescription ID	service-specific
IPDE0149	Presenting Concerns	service-specific
IPDE0150	Pressure Injury Risk	service-specific
IPDE0151	Procedure Clinician	service-specific
IPDE0152	Procedure Code	service-specific
IPDE0153	Procedure Date Time	service-specific
IPDE0154	Procedure ID	service-specific
IPDE0155	Procedure Name	service-specific
IPDE0156	Procedure Notes	service-specific
IPDE0157	Procedure Outcome	service-specific
IPDE0158	Procedures Summary	service-specific
IPDE0159	Progress Note	service-specific
IPDE0160	Progress SOAP Assessment	service-specific
IPDE0161	Progress SOAP Objective	service-specific
IPDE0162	Progress SOAP Plan	service-specific
IPDE0163	Progress SOAP Subjective	service-specific
IPDE0164	Proposed Admission Type	service-specific
IPDE0165	Pulse Per Minute	service-specific
IPDE0166	Quantity Issued	service-specific
IPDE0167	Quantity Ordered	service-specific
IPDE0168	Readmission Reference	service-specific
IPDE0169	Reason for Admission	service-specific
IPDE0170	Reason for Consultation	service-specific
IPDE0171	Receiving Contact	service-specific
IPDE0172	Receiving Facility	service-specific

DER Code	Metadata name	Scope
IPDE0173	Receiving Staff	service-specific
IPDE0174	Recommendations	service-specific
IPDE0175	Reconciliation Reason	service-specific
IPDE0176	Recorded By	service-specific
IPDE0177	Recovery Disposition	service-specific
IPDE0178	Recovery Score	service-specific
IPDE0179	Recovery Start Time	service-specific
IPDE0180	Reference Range	service-specific
IPDE0181	Referral ID	service-specific
IPDE0182	Referral Priority	service-specific
IPDE0183	Referral Reason	service-specific
IPDE0184	Referral Status	service-specific
IPDE0185	Referred From	service-specific
IPDE0186	Report Findings	service-specific
IPDE0187	Report Impression	service-specific
IPDE0188	Reporting Clinician	service-specific
IPDE0189	Request Time	service-specific
IPDE0190	Requested Ward	service-specific
IPDE0191	Respiratory Rate Per Minute	service-specific
IPDE0192	Response to Care	service-specific
IPDE0193	Responsible Clinician	service-specific
IPDE0194	Responsible Nurse	service-specific
IPDE0195	Result Interpretation	service-specific
IPDE0196	Result Time	service-specific
IPDE0197	Result Unit	service-specific
IPDE0198	Result Value	service-specific
IPDE0199	Reviewed By	service-specific
IPDE0200	Round Time	service-specific
IPDE0201	Route	service-specific
IPDE0202	Safety Checklist Complete	service-specific
IPDE0203	Sample Type	service-specific
IPDE0204	Scheduled Dose Time	service-specific
IPDE0205	Secondary Diagnoses	service-specific
IPDE0206	Service Category	service-specific
IPDE0207	Session Date Time	service-specific
IPDE0208	Shift	service-specific
IPDE0209	Specialty	service-specific
IPDE0210	Specimen Condition	service-specific
IPDE0211	Specimen ID	service-specific
IPDE0212	Start Date Time	service-specific
IPDE0213	Study Date Time	service-specific
IPDE0214	Study Reference	service-specific
IPDE0215	Surgeon ID	service-specific
IPDE0216	Surgery End Time	service-specific

DER Code	Metadata name	Scope
IPDE0217	Surgery ID	service-specific
IPDE0218	Surgery Start Time	service-specific
IPDE0219	Surgical History	service-specific
IPDE0220	Surgical Procedure	service-specific
IPDE0221	Systolic BP	service-specific
IPDE0222	Temperature Celsius	service-specific
IPDE0223	Test Code	service-specific
IPDE0224	Test Name	service-specific
IPDE0225	Test Order ID	service-specific
IPDE0226	To Bed	service-specific
IPDE0227	To Ward	service-specific
IPDE0228	Total Volume ML	service-specific
IPDE0229	Transfer ID	service-specific
IPDE0230	Transfer Reason	service-specific
IPDE0231	Transfer Time	service-specific
IPDE0232	Transfusion End	service-specific
IPDE0233	Transfusion Order ID	service-specific
IPDE0234	Transfusion Outcome	service-specific
IPDE0235	Transfusion Start	service-specific
IPDE0236	Treatment Summary	service-specific
IPDE0237	Verified By	service-specific
IPDE0238	Waiver or Exemption	service-specific
IPDE0239	Witnessed By	service-specific
IPDE0240	Workflow Status	service-specific
IPDE0241	Working Diagnosis	service-specific
IVDE0001	INV - Adjustment Approval	service-specific
IVDE0002	INV - Approved By	service-specific
IVDE0003	INV - Authorisation Reference	service-specific
IVDE0004	INV - Base Unit	service-specific
IVDE0005	INV - Batch or Lot Number	service-specific
IVDE0006	INV - Consumption Context	service-specific
IVDE0007	INV - Count Date	service-specific
IVDE0008	INV - Count Reference	service-specific
IVDE0009	INV - Counted Quantity	service-specific
IVDE0010	INV - Destination Stock Location	service-specific
IVDE0011	INV - Dispatch Reference	service-specific
IVDE0012	INV - Disposal Method	service-specific
IVDE0013	INV - Expected Delivery	service-specific
IVDE0014	INV - Expiry Date	service-specific
IVDE0015	INV - Goods Received Note	service-specific
IVDE0016	INV - Linked Order ID	service-specific
IVDE0017	INV - Linked Visit ID	service-specific
IVDE0018	INV - Opening Quantity	service-specific
IVDE0019	INV - Ordered Quantity	service-specific

DER Code	Metadata name	Scope
IVDE0020	INV - Patient Reference Token	service-specific
IVDE0021	INV - Performed By User ID	service-specific
IVDE0022	INV - Prescription ID	service-specific
IVDE0023	INV - Purchase Order Reference	service-specific
IVDE0024	INV - Quality Assessment	service-specific
IVDE0025	INV - Quantity	service-specific
IVDE0026	INV - Reason	service-specific
IVDE0027	INV - Requested Quantity	service-specific
IVDE0028	INV - Requesting Service Point	service-specific
IVDE0029	INV - Required By	service-specific
IVDE0030	INV - Requisition ID	service-specific
IVDE0031	INV - Reservation Expiry	service-specific
IVDE0032	INV - Reservation ID	service-specific
IVDE0033	INV - Reserved Quantity	service-specific
IVDE0034	INV - Return Source	service-specific
IVDE0035	INV - Service Event ID	service-specific
IVDE0036	INV - Source Operation ID	service-specific
IVDE0037	INV - Source Stock Location	service-specific
IVDE0038	INV - Stock Location	service-specific
IVDE0039	INV - Supplier Name	service-specific
IVDE0040	INV - System Quantity	service-specific
IVDE0041	INV - Transaction ID	service-specific
IVDE0042	INV - Transaction Status	service-specific
IVDE0043	INV - Transaction Time	service-specific
IVDE0044	INV - Transaction Type	service-specific
IVDE0045	INV - Transfer Reference	service-specific
IVDE0046	INV - Unit Cost	service-specific
LBDE0001	Analyzer reference	cross-cutting
LBDE0002	Collection date and time	cross-cutting
LBDE0003	Critical result alert	cross-cutting
LBDE0004	Notification time	cross-cutting
LBDE0005	Notified clinician ID	cross-cutting
LBDE0006	Order status	cross-cutting
LBDE0007	Quality control result	cross-cutting
LBDE0008	Reference interval	cross-cutting
LBDE0009	Result authorised by	cross-cutting
LBDE0010	Result units	cross-cutting
LBDE0011	Result value text	cross-cutting
LBDE0012	Specimen reference	cross-cutting
LBDE0013	Specimen status	cross-cutting
LBDE0014	Test catalogue code	cross-cutting
MDDE0001	Prescription reference	cross-cutting
MDDE0002	Treatment regimen code	cross-cutting
MHDE0001	Mental health risk	service-specific

DER Code	Metadata name	Scope
MHDE0002	Mental state examination	service-specific
MHDE0003	Presenting concern	service-specific
MHDE0004	Suicide risk note	service-specific
MHDE0005	Therapy session summary	service-specific
MTDE0001	Birth date and time	service-specific
MTDE0002	Birth mode	service-specific
MTDE0003	Birth outcome	service-specific
MTDE0004	Cervical dilatation cm	service-specific
MTDE0005	Contraction frequency	service-specific
MTDE0006	Discharge date	service-specific
MTDE0007	Estimated blood loss (mL)	service-specific
MTDE0008	Fetal heart rate	service-specific
MTDE0009	Intervention indication	service-specific
MTDE0010	Labour onset time	service-specific
MTDE0011	Maternal disposition	service-specific
MTDE0012	Maternal risk note	service-specific
MTDE0013	Newborn linked patient identifier	service-specific
MTDE0014	Postpartum complication	service-specific
NCDE0001	Cardiovascular risk factors	service-specific
NCDE0002	Diabetes type	service-specific
NCDE0003	Fasting glucose (mmol/L)	service-specific
NCDE0004	Foot exam result	service-specific
NCDE0005	HbA1c (%)	service-specific
NCDE0006	Hypertension diagnosis	service-specific
NCDE0007	Renal risk note	service-specific
NCDE0008	Retinal screening result	service-specific
NCDE0009	Urine albumin result	service-specific
NUDE0001	Diet plan reference	service-specific
NUDE0002	Nutrition response	service-specific
ONDE0001	Cycle number	service-specific
ONDE0002	Decision date	service-specific
ONDE0003	Diagnosis date	service-specific
ONDE0004	Histology terminology code	service-specific
ONDE0005	Palliative intervention	service-specific
ONDE0006	Primary cancer site code	service-specific
ONDE0007	Protocol code	service-specific
ONDE0008	Radiotherapy course reference	service-specific
ONDE0009	Stage at diagnosis	service-specific
ONDE0010	Treatment administration date	service-specific
ONDE0011	Treatment response code	service-specific
OPDE0001	Adverse Event	service-specific
OPDE0002	Billing Balance UGX	service-specific
OPDE0003	Billing Clearance Date Time	service-specific
OPDE0004	Billing Clearance Granted	service-specific

DER Code	Metadata name	Scope
OPDE0005	Billing Clearance Reason	service-specific
OPDE0006	Charge Description	service-specific
OPDE0007	Charge Discount Amount UGX	service-specific
OPDE0008	Charge Gross Amount UGX	service-specific
OPDE0009	Charge Line Reference	service-specific
OPDE0010	Charge Net Amount UGX	service-specific
OPDE0011	Charge Quantity	service-specific
OPDE0012	Charge Service Category	service-specific
OPDE0013	Charge Source Reference	service-specific
OPDE0014	Charge Status	service-specific
OPDE0015	Charge Unit Price UGX	service-specific
OPDE0016	Check-in Time	service-specific
OPDE0017	Clinical Examination	service-specific
OPDE0018	Clinical Outcome	service-specific
OPDE0019	Clinical Plan	service-specific
OPDE0020	Collection Date Time	service-specific
OPDE0021	Critical Result Escalated	service-specific
OPDE0022	Diastolic Blood Pressure	service-specific
OPDE0023	Discharge Summary	service-specific
OPDE0024	Dispensing Date Time	service-specific
OPDE0025	Dispensing Notes	service-specific
OPDE0026	Disposition	service-specific
OPDE0027	Dose Unit	service-specific
OPDE0028	Duration Days	service-specific
OPDE0029	External Transaction Reference	service-specific
OPDE0030	Final Diagnosis	service-specific
OPDE0031	Follow-up Date	service-specific
OPDE0032	Follow-up Findings	service-specific
OPDE0033	Frequency	service-specific
OPDE0034	Heart Rate	service-specific
OPDE0035	Height Cm	service-specific
OPDE0036	History of Present Illness	service-specific
OPDE0037	Identity Verified	service-specific
OPDE0038	Imaging Body Site	service-specific
OPDE0039	Imaging Clinical Indication	service-specific
OPDE0040	Imaging Findings	service-specific
OPDE0041	Imaging Impression	service-specific
OPDE0042	Imaging Modality	service-specific
OPDE0043	Imaging Performed Date Time	service-specific
OPDE0044	Invoice Reference	service-specific
OPDE0045	Laboratory Reference Range	service-specific
OPDE0046	Laboratory Report Date Time	service-specific
OPDE0047	Laboratory Result Interpretation	service-specific
OPDE0048	Laboratory Result Units	service-specific

DER Code	Metadata name	Scope
OPDE0049	Laboratory Result Value	service-specific
OPDE0050	Laboratory Test Name	service-specific
OPDE0051	Laboratory Test Requested	service-specific
OPDE0052	Medicine Dispensed	service-specific
OPDE0053	Medicine Prescribed	service-specific
OPDE0054	Order Priority	service-specific
OPDE0055	Order Reference	service-specific
OPDE0056	Oxygen Saturation Percent	service-specific
OPDE0057	Payer Category	service-specific
OPDE0058	Payment Amount UGX	service-specific
OPDE0059	Payment Date Time	service-specific
OPDE0060	Payment Method	service-specific
OPDE0061	Payment Notes	service-specific
OPDE0062	Payment Reference	service-specific
OPDE0063	Payment Status	service-specific
OPDE0064	Prescribed Dose	service-specific
OPDE0065	Procedure Date Time	service-specific
OPDE0066	Procedure Name	service-specific
OPDE0067	Procedure Outcome	service-specific
OPDE0068	Provisional Diagnosis	service-specific
OPDE0069	Quantity Dispensed	service-specific
OPDE0070	Quantity Prescribed	service-specific
OPDE0071	Queue Number	service-specific
OPDE0072	Reason for Visit	service-specific
OPDE0073	Referral Date	service-specific
OPDE0074	Referral Destination	service-specific
OPDE0075	Referral Reason	service-specific
OPDE0076	Referral Status	service-specific
OPDE0077	Referral Type	service-specific
OPDE0078	Registration Notes	service-specific
OPDE0079	Request Indication	service-specific
OPDE0080	Request Status	service-specific
OPDE0081	Requested Service	service-specific
OPDE0082	Respiratory Rate	service-specific
OPDE0083	Return Precautions	service-specific
OPDE0084	Route of Administration	service-specific
OPDE0085	Specimen Identifier	service-specific
OPDE0086	Specimen Notes	service-specific
OPDE0087	Specimen Received Date Time	service-specific
OPDE0088	Specimen Type	service-specific
OPDE0089	Systolic Blood Pressure	service-specific
OPDE0090	Temperature Celsius	service-specific
OPDE0091	Triage Category	service-specific
OPDE0092	Triage Complaint	service-specific

DER Code	Metadata name	Scope
OPDE0093	Visit Payment Category	service-specific
OPDE0094	Visit Reference	service-specific
OPDE0095	Weight Kg	service-specific
PNDE0001	Breastfeeding counselling	service-specific
PNDE0002	Delivery date	service-specific
PNDE0003	Delivery facility	service-specific
PNDE0004	Jaundice assessment	service-specific
PNDE0005	Mother linked patient identifier	service-specific
PNDE0006	Newborn danger signs	service-specific
RDDE0001	Appointment date	service-specific
RDDE0002	Body site code	service-specific
RDDE0003	Contrast administered	service-specific
RDDE0004	Imaging study identifier	service-specific
RDDE0005	Modality code	service-specific
RDDE0006	Radiology report	service-specific
RDDE0007	Reported date	service-specific
RFDE0001	Referral reference	cross-cutting
RHDE0001	Discharge recommendation	service-specific
RHDE0002	Functional baseline scale	service-specific
RHDE0003	Functional outcome score	service-specific
RHDE0004	Session duration minutes	service-specific
RHDE0005	Therapy goal	service-specific
RHDE0006	Therapy intervention code	service-specific
RNDE0001	Baseline renal function note	service-specific
RNDE0002	Complication note	service-specific
RNDE0003	Dialysis duration minutes	service-specific
RNDE0004	Dialysis session status	service-specific
RNDE0005	Post dialysis weight kg	service-specific
RNDE0006	Pre dialysis weight kg	service-specific
RNDE0007	Renal diagnosis code	service-specific
RNDE0008	Vascular access type	service-specific
SUDE0001	Anaesthesia risk classification	service-specific
SUDE0002	Operative note	service-specific
SUDE0003	Planned procedure code	service-specific
SUDE0004	Postoperative complication	service-specific
SUDE0005	Preoperative findings	service-specific
SUDE0006	Procedure end time	service-specific
SUDE0007	Procedure start time	service-specific
SUDE0008	Sign-in completed	service-specific
SUDE0009	Sign-out completed	service-specific
SUDE0010	Surgery date	service-specific
SUDE0011	Time-out completed	service-specific
SUDE0012	Wound classification	service-specific
TBDE0001	Contact patient identifier	service-specific

DER Code	Metadata name	Scope
TBDE0002	Disease classification	service-specific
TBDE0003	Index case reference	service-specific
TBDE0004	Molecular test result	service-specific
TBDE0005	Screening result	service-specific
TBDE0006	Smear result	service-specific
TBDE0007	Symptoms duration days	service-specific
TBDE0008	TB outcome	service-specific
TBDE0009	Treatment phase	service-specific
TBDE0010	Treatment start date	service-specific
VSDE0001	Birth weight (g)	cross-cutting
VSDE0002	Diastolic BP (mmHg)	cross-cutting
VSDE0003	Height (cm)	cross-cutting
VSDE0004	MUAC (mm)	cross-cutting
VSDE0005	Systolic BP (mmHg)	cross-cutting
VSDE0006	Weight (kg)	cross-cutting
WFDE0001	Encounter reference	cross-cutting
WFDE0002	Order reference	cross-cutting

programRules

DER Code	Metadata name	Scope
ANPR0001	Gestational age weeks warning	service-specific
ANPR0002	Systolic BP (mmHg) warning	service-specific
ERPR0001	Heart rate (per min) warning	service-specific
ERPR0002	Oxygen Saturation (SpO2) (%) warning	service-specific
IMPR0001	Weight (kg) warning	service-specific
NCPR0001	Glycated Haemoglobin (HbA1c) (%) warning	service-specific
NCPR0002	Systolic BP (mmHg) warning	service-specific
RNPR0001	Dialysis duration minutes warning	service-specific

programIndicators

DER Code	Metadata name	Scope
ANPI0001	Antenatal Care - Recorded Events	service-specific
BBPI0001	Blood Bank and Transfusion - Recorded Events	service-specific
DNPI0001	Dental and Oral Health - Recorded Events	service-specific
ENPI0001	Ear, Nose and Throat Services - Recorded Events	service-specific
EQPI0001	Facility Equipment and Maintenance - Recorded Events	service-specific
ERPI0001	Emergency and Resuscitation - Recorded Events	service-specific
EYPI0001	Eye and Vision Care - Recorded Events	service-specific
FPPI0001	Family Planning and Reproductive Health - Recorded Events	service-specific
HVPI0001	HIV Care and Antiretroviral Therapy - Recorded Events	service-specific
ICPI0001	Hospital Infection Prevention and Control - Recorded Events	service-specific
IMPI0001	Child Health and Immunisation - Recorded Events	service-specific
IPPI0001	Integrated Inpatient EMR - Recorded Events	service-specific
IVPI0001	Inventory and Supplies Tracker - Recorded Events	service-specific
LBPI0001	Laboratory Diagnostics - Recorded Events	cross-cutting
MHPI0001	Mental Health and Psychiatry - Recorded Events	service-specific
MTPI0001	Labour and Delivery Care - Recorded Events	service-specific

DER Code	Metadata name	Scope
NCPI0001	Diabetes and Hypertension Care - Recorded Events	service-specific
NUPI0001	Nutrition and Dietetics - Recorded Events	service-specific
ONPI0001	Cancer and Oncology - Recorded Events	service-specific
OPPI0001	Integrated Outpatient EMR - Recorded Events	service-specific
PNPI0001	Postnatal and Newborn Follow-up - Recorded Events	service-specific
RDPI0001	Imaging and Radiology - Recorded Events	service-specific
RHPI0001	Rehabilitation and Physiotherapy - Recorded Events	service-specific
RNPI0001	Renal Care and Dialysis - Recorded Events	service-specific
SUPI0001	Surgical and Theatre Services - Recorded Events	service-specific
TBPI0001	Tuberculosis and Leprosy Care - Recorded Events	service-specific